

SAVITRIBAI PHULE PUNE UNIVERSITY



**A PROJECT STAGE II REPORT
ON**

**Nirikshak: An Industrial IoT & Predictive Maintenance
Framework**

**SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF**

**BACHELOR OF ENGINEERING
BY**

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**DEPARTMENT OF ELECTRONICS AND
TELECOMMUNICATION
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Abstract

Predictive maintenance is an essential technology in Industry 4.0 for increasing equipment reliability, reducing unplanned downtime and minimizing maintenance costs. Conventional maintenance methods are not based on continuous condition monitoring and often lead to unnecessary maintenance or unexpected equipment failure. To tackle these challenges, this project proposes **Nirikshak: An Industrial IoT & Predictive Maintenance Framework** which provides a low cost and scalable solution for real-time machine health monitoring and intelligent fault prediction.

The proposed framework uses a Data Acquisition (DAQ) device based on the STM32F103 integrated with temperature, vibration, RPM, voltage and current sensors to monitor continuously machine operating conditions. The collected sensor data is sent to an edge computing system via an ESP32 Wi-Fi module using the MQTT communication protocol. A Python-centric processing layer is employed for data preprocessing, feature extraction, anomaly detection, fault classification and Remaining Useful Life (RUL) estimation.

Machine learning techniques are used to provide predictive maintenance intelligence. We used Isolation Forest for the anomaly detection to detect abnormal machine behavior, Random Forest Classifier for the fault classification and Random Forest Regressor for Remaining Useful Life prediction. The processed sensor data and prediction results are synchronized with Firebase Realtime Database and visualized through a web-based dashboard that provides real time monitoring, machine health status, alerts, and historical trend analysis.

The developed framework was tested with machine condition data obtained under different operating conditions. Experimental results show effective anomaly detection, accurate fault classification, reliable RUL estimation, low communication latency and real-time responsiveness. The total hardware implementation cost is around Rs.8,500 which makes the system affordable and useful for small and medium scale industries. The proposed framework offers a feasible, scalable, and Industry 4.0-enabled solution for intelligent predictive maintenance applications.

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CHAPTER 1

INTRODUCTION

1.1 Overview

The rapid adoption of Industry 4.0 has developed intelligent maintenance solutions capable of improving equipment reliability and reducing operational downtime. Predictive Maintenance (PdM) has emerged as an effective maintenance strategy that utilizes real-time machine condition data and advanced analytics for earlier fault detection. Unlike conventional reactive and preventive maintenance approaches, predictive maintenance enables condition-based decision-making, thereby reducing maintenance costs and improving asset utilization.

The proposed project, Nirikshak: An Industrial IoT-Based Predictive Maintenance Framework, integrates embedded systems, Industrial Internet of Things (IIoT), edge computing, cloud computing, and machine learning into a unified monitoring platform. The system employs an STM32-based Data Acquisition (DAQ) unit interfaced with temperature, vibration, RPM, voltage, and current sensors to continuously monitor machine health parameters. The collected data is transmitted using the MQTT protocol and processed at an edge computing node for feature extraction and analysis. The collected sensor information is analyzed using intelligent prediction models that identify equipment faults, detect abnormal behavior, and estimate the remaining operational lifespan of machinery. Firebase Realtime Database act as a cloud based platform for data storage and synchronization, while a web-based dashboard provides real-time visualization of machine conditions, alerts, and predictive insights. The framework offers a low-cost, scalable, and practical solution for modern industrial maintenance applications.

1.2 Need and Relevance

Industrial equipment failures often result in unplanned downtime, production losses, increased maintenance expenses, and reduced operational efficiency. Traditional maintenance strategies either react after a failure occurs or follow fixed maintenance schedules without considering the actual condition of the machine. Such approaches can lead to inefficient resource utilization and unexpected breakdowns.

The growing adoption of Industry 4.0 technologies identifies the need for intelligent maintenance systems capable of continuous monitoring and predictive analysis. An Industrial IoT-based predictive maintenance helps to detect faults early through real-time sensor data and machine learning techniques. Small and medium-sized industries mostly used this proposed system, by combining low-cost hardware, cloud connectivity, and advanced analytics to improve machine reliability and maintenance planning.

1.3 Motivation

Modern industries increasingly rely on uninterrupted machine operation to maintain productivity and competitiveness. Mostly, predictive maintenance solutions are expensive, proprietary, and difficult to customize for specific industrial requirements. These limitations restrict their adoption, especially among small and medium enterprises.

The motivation behind this project is to develop an accessible and cost-effective predictive maintenance framework that combines Industrial IoT technologies with machine learning-based intelligence. By integrating real-time sensing, edge computing, cloud storage, and predictive analytics, the system provides actionable insights into machine health conditions. The ability to detect anomalies, predict faults, and estimate Remaining Useful Life can help industries minimize downtime, reduce maintenance costs, and improve operational safety. Furthermore, the project demonstrates the practical implementation of Industry 4.0 concepts using open-source technologies and low-cost embedded hardware.

1.4 Problem Statement

To develop a low-cost, customizable Industrial IoT system that collects real-time machine data, predicts failures using machine learning, and displays key parameters through a user-friendly dashboard for efficient predictive maintenance.

1.5 Aim and Objectives

Industrial IoT-based predictive maintenance framework is capable of real-time machine monitoring, fault prediction, anomaly detection, and Remaining Useful Life estimation using embedded sensing, edge computing, cloud integration, and machine learning techniques

Objectives:

- To make the DAQ (Data Acquisition) device efficiently collect sensor data in real time.
- Real time transmission of collected data to a cloud platform running on a processing device.
- To implement proper data processing and machine learning techniques for predicting potential failures.

- To develop a custom dashboard that easily interprets data and sends notifications and alerts for timely action.
- To estimate machine health status and Remaining Useful Life (RUL) using predictive analytics techniques.

1.6 Project Scope and Limitations

The proposed framework focuses on real-time monitoring and predictive maintenance of industrial machines such as motors, pumps, compressors, and other rotating equipment. The system supports multi-sensor data acquisition, MQTT-based communication, edge-level data processing, cloud synchronization through Firebase, machine learning-based fault prediction, anomaly detection, and RUL estimation. The modular architecture allows future scalability through the addition of new sensors, machines, and predictive models.

However, the system is developed and validated in a controlled experimental environment using a limited set of machine operating conditions and sensor configurations. The accuracy of fault prediction depends on available training data. Large-scale industrial deployment may require additional cybersecurity measures, model retraining mechanisms, and integration with existing industrial automation infrastructure. Furthermore, environmental variations and unforeseen machine faults may affect prediction performance.

1.7 Organisation of the Report

The project report is organized as follows:

Chapter 1 introduces the Nirikshak framework as a solution and describes the overview, need and relevance, motivation, problem statement, objective, scope and limitation of the project.

Chapter 2 covers literature survey on Industrial IoT, predictive maintenance, machine learning, edge computing and other relevant researches.

Chapter 3 describes the proposed methodology, project framework, data acquisition process, communication architecture, and overall system workflow.

Chapter 4 explains the design workflow and implementation of the proposed system, including the system architecture, flowchart, hardware components, sensor interfacing, and PCB development.

Chapter 5 discusses the software implementation of the framework, covering MQTT communication, data preprocessing, machine learning models, cloud integration, and dashboard visualization.

Chapter 6 presents the testing methodology, troubleshooting procedures, testing results, and performance analysis of the developed system.

Chapter 7 highlights the advantages and industrial applications of the proposed predictive maintenance framework.

Chapter 8 concludes the report by summarizing the major outcomes of the project and discusses the future scope improvements and large-scale deployment.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Survey

Predictive maintenance is one of the prominent technologies of Industry 4.0 for which various research papers, journals, conference papers, and industrial papers on IIoT, machine condition monitoring, anomaly detection, fault diagnosis and RUL prediction have been surveyed. The surveyed literature demonstrates how the combination of IoT devices, cloud computing, edge computing, artificial intelligence, and machine learning algorithms is used to design intelligent predictive maintenance systems. This review discusses some of the important contributions made by the recent research work related to the proposed Nirikshak: An Industrial IoT Predictive Maintenance Framework. The papers selected for this survey were identified from references and related papers of the project research work..

1. Theissler et al. (2021) [1], in their paper "Machine Learning Techniques for Predictive Maintenance in the Auto- motive Industry" published in Reliability Engineering System Safety was an in-depth review of the use of machine learning techniques in predicting maintenance needs in industry. This article covered the use of machine learning techniques in condition monitoring, failure diagnosis and maintenance planning based on data. Machine learning techniques, sensor technology, and industrial data analytics were applied. The results proved that machine learning can increase the accuracy of failure prediction and enhance maintenance efficiency. Nevertheless, issues like data quality and deployment were encountered.
2. Samatas et al. (2021) [2], in their paper "Predictive Maintenance: Bridging Artificial Intelligence and IoT" presented at the IEEE World AI IoT Congress (AIIoT), proposed an integrated framework combining Artificial Intelligence and Internet of Things technologies for predictive maintenance applications. The study focused on continuous equipment monitoring through connected sensors and intelligent analytics. The work utilized IoT communication systems, cloud computing platforms, and AI-based prediction models. The results demonstrated improved decision-making capabilities and proactive maintenance scheduling. The proposed approach contributed to reducing unexpected equipment failures and operational downtime.
3. Tasci et al. (2023)[3] in their paper "Remaining Useful Lifetime Prediction for Predictive Maintenance in Manufacturing" published in Computers & Industrial Engineering, developed a framework for estimating Remaining Useful Life (RUL) of industrial equipment. The research focused on prognostics and

health management using historical and real-time machine data. Technologies such as time-series analysis, machine learning algorithms, and predictive analytics were employed. The study demonstrated that accurate RUL prediction enables optimized maintenance planning and better resource utilization. The system improved operational reliability by providing early warnings before equipment failure.

4. Savaglio et al. (2024) [4] in their paper “Edge Intelligence for Industrial IoT: Opportunities and Limitations” published in *Procedia Computer Science*, discussed the role of edge computing in Industrial IoT applications. The study explored how intelligence can be deployed closer to industrial machines for faster decision-making and reduced network latency. Technologies including edge computing, Industrial IoT platforms, distributed analytics, and machine learning were investigated. The findings demonstrated that edge intelligence improves real-time responsiveness and bandwidth efficiency. However, limitations related to computational resources and security challenges were also highlighted.
5. Elkateb et al. (2024) [5], in their paper “Machine Learning and IoT-Based Predictive Maintenance Approach for Industrial Applications” published in the *Alexandria Engineering Journal*, proposed an intelligent predictive maintenance system using IoT-enabled sensors and machine learning models. The work focused on monitoring equipment health and detecting abnormal operating conditions at an early stage. Technologies such as IoT sensors, cloud computing, supervised learning algorithms, and data analytics were utilized. The results demonstrated improved fault detection performance and enhanced maintenance efficiency. The study emphasized the importance of integrating machine learning with real-time industrial monitoring systems.
6. Cummins et al. (2024)[6] in their paper “Explainable Predictive Maintenance: A Survey of Current Methods, Challenges and Opportunities” published in *IEEE Access*, reviewed explainable artificial intelligence techniques used in predictive maintenance applications. The study focused on improving transparency and trust in machine learning-based maintenance systems. Technologies including Explainable AI (XAI), deep learning, predictive analytics, and industrial monitoring systems were analyzed. The findings demonstrated that explainable models improve user confidence and facilitate maintenance decision-making. The paper also identified challenges associated with balancing model accuracy and interpretability.

7. Andriulo et al. (2024)[7] in their paper “Edge Computing and Cloud Computing for Internet of Things: A Review” published in Informatics, presented a detailed review of edge-cloud architectures for IoT applications. The research focused on efficient data processing, storage, and communication mechanisms for large-scale IoT systems. Technologies such as cloud computing, edge computing, IoT frameworks, and distributed systems were examined. The study showed that combining edge and cloud platforms improves scalability, reliability, and real-time processing capabilities. The proposed concepts are highly relevant for industrial predictive maintenance deployments.
8. Artiushenko et al. (2024)[8] in their paper “Resource-Efficient Edge AI Solution for Predictive Maintenance” published in Procedia Computer Science, proposed a lightweight Edge AI framework for industrial predictive maintenance. The study focused on deploying machine learning models directly on resource-constrained edge devices. Technologies such as Edge AI, embedded systems, sensor networks, and optimized machine learning algorithms were employed. The results demonstrated reduced latency and efficient fault detection without excessive cloud dependency. The framework highlighted the feasibility of low-cost intelligent maintenance systems for industrial applications.
9. Hamasha et al. (2025)[9] in their paper “A Comprehensive Framework for IoT-Driven Predictive Maintenance: Leveraging AI and Edge Computing for Enhanced Equipment Reliability” published in the Journal of Applied Engineering Science, proposed a complete predictive maintenance architecture integrating IoT, Artificial Intelligence, and edge computing. The work focused on improving equipment reliability through continuous monitoring and intelligent analytics. Technologies including IoT sensors, edge devices, machine learning models, and cloud infrastructure were utilized. The findings demonstrated significant improvements in fault prediction accuracy and maintenance efficiency. The framework provided a scalable solution suitable for Industry 4.0 environments
10. Rakholia et al. (2025) [10] in their paper “Integrating AI and IoT for Predictive Maintenance in Industry 4.0 Manufacturing Environments: A Practical Approach” published in Information, presented a practical implementation of AI-enabled predictive maintenance for smart manufacturing systems. The study focused on real-time machine monitoring, anomaly detection, and predictive analytics. Technologies such as Industrial IoT, machine learning, cloud com-

puting, and intelligent sensors were employed. The proposed approach improved equipment availability and reduced maintenance costs through proactive fault management. The study demonstrated the effectiveness of combining AI and IoT technologies for industrial applications.

2.2 Research Gap

It can be concluded from the literature review that the application of predictive maintenance with Industrial IoT has become an interesting field of research because of using machine learning, edge computing, and cloud computing technologies. The majority of the studies prove that machine learning has already been used successfully in fault detection, anomaly detection, and condition monitoring. However, there is an evident shortage of comprehensive and practical solutions in the area of predictive maintenance. In most cases, existing solutions cover only separate elements of the process.

Most of the developed approaches are based on the cloud processing, which is likely to cause higher latencies and non-real time processing problems. Also, some of the studies consider only the prediction algorithms without combining multi-sensor data collection, communication, edge computing, cloud storage and visualization. In addition, some of the industrial predictive maintenance solutions need expensive software and hardware infrastructure to implement the solution.

Thus, such solutions are less applicable for small and medium-sized enterprises. Other limitation that exists in the literature is absence of low-cost systems which integrate fault classification, anomaly detection and RUL prediction. In addition to that there is an absence of real-life deployment implementations that use affordable hardware with real time monitoring capabilities. To solve the above stated limitations, the proposed NIRIKSHAK: An Industrial IoT Predictive Maintenance Framework brings together multiple sensor monitoring, MQTT based communication, edge and cloud computing along with machine learning approaches in one single architecture. The system makes use of Random Forest Classifier for fault classification, Isolation Forest for anomaly detection and Random Forest Regressor for RUL prediction.

2.2.1 Research Gaps Identified

The following research gaps were identified from the literature survey:

- Lack of complete end-to-end IIoT-based predictive maintenance frameworks.
- Limited integration of fault classification, anomaly detection, and Remaining Useful Life (RUL) estimation within a single framework.

- Heavy dependence on cloud-based processing, resulting in increased latency and reduced real-time responsiveness.
- High implementation cost of existing solutions, making them difficult for Small and Medium-sized Enterprises (SMEs) to adopt.
- Limited utilization of edge computing for real-time data processing and analytics.
- Lack of affordable, scalable, and practically deployable industrial predictive maintenance systems.
- Insufficient integration of real-time monitoring dashboards with predictive maintenance models for effective decision-making.

2.3 Summary

In Industrial 4.0 settings, predictive maintenance is evident through an extensive body of literature. Studies have shown how Industrial IoT, machine-learning algorithms, edge computing, cloud platforms, anomaly detection, and RUL predictions improve machine reliability and decrease downtime. As such, these findings inspired the creation of the Nirikshak framework; a complete predictive maintenance solution that leverages real-time monitoring, MQTT communication, cloud connectivity, and machine-learning predictive analytics into one cohesive system.

CHAPTER 3

METHODOLOGY

3.1 Project Framework

Nirikshak: An Industrial Iot & Predictive Maintenance Framework will be designed to monitor the condition of industrial machines and predict possible failure beforehand. The proposed framework combines IIoT, embedded systems, machine learning, cloud storage, and visualization to offer an intelligent maintenance solution.

The system utilizes an edge-cloud computing architecture wherein condition parameters of the machine including temperature, vibrations, voltage, current, and revolutions per minute are obtained through various sensors and DAQ system, namely STM32. The data obtained are then sent wirelessly using an ESP32 communication module and MQTT protocol to an edge computing system.

In terms of edge computing, machine learning algorithms are used for analyzing and processing the data for detecting faults, identifying anomalies, determining the machine's health condition, and scheduling preventive maintenance tasks. The processed data and the prediction results are synced with the Firebase Real-time Database and visualized via the web interface.

3.2 Block Diagram

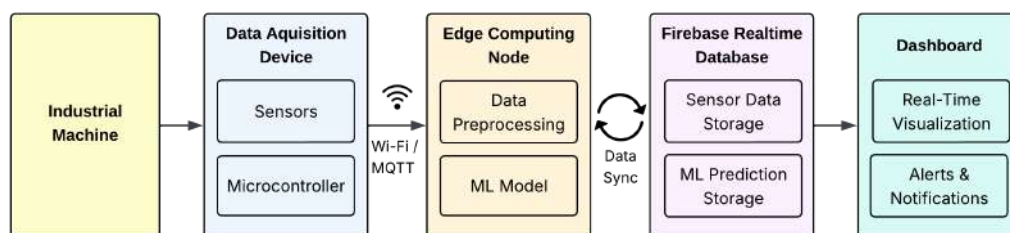


Figure 3.1: Block Diagram

3.2.1 Components Of Block Diagram:

1. Industrial Machine

The industrial machinery means the machines being observed through the process of predictive maintenance. Machines like motors, pumps, compressors, and rotating machines generate various operational data such as temperature, vibrations, voltage, current, and speed. Continuous observation of such parameters allows us to evaluate the health status of the machines and detect any anomalies.

2. Data Acquisition Device

The Data Acquisition Module is made up of sensors which are connected to the STM32F103 microcontroller and ESP32 wireless module. These sensors continually collect data concerning the performance of the machine including its temperature, vibrations, voltage, current, and RPM.

- DS18B20 Temperature Sensor – Measures machine temperature.
- MPU6050 Sensor – Monitors vibration characteristics.
- Inductive Proximity Sensor – Measures shaft rotational speed (RPM).
- ZMPT101B Voltage Sensor – Measures supply voltage.
- SCT-013 Current Sensor – Measures current consumption.

STM32F103 microcontroller is responsible for data collection, ADC conversion, data validation, and packaging the collected data for transmission. The ESP32 wireless module acts as a Wi-Fi communication node and transfers the collected sensor data to the edge computing node through the MQTT protocol.

3. Edge Computing Node

Data is obtained in real-time through MQTT and processed locally at the Edge Computing Node. Preprocessing includes activities such as data cleaning, normalization, and extraction of features.

Machine learning techniques are applied to determine the machine's condition, detect any deviations or anomalies, classify any faults that arise, and estimate potential future degradations.

4. Firebase Realtime Database

The Firebase Realtime Database is the cloud-based storage solution for the system. This database contains sensor readings, ML predictions, anomaly detection data, and logs. The database allows for centralization of data management and enables remote access to the data.

5. Dashboard and Visualization Layer

Dashboard provides the operator with an intuitive interface for monitoring equipment in real time. It pulls data from Firebase Realtime Database to display live readings from sensors, the condition of the equipment, prediction results, trends in data and maintenance alerts. Dashboard helps the operator to make well-informed maintenance decisions to minimize the frequency of equipment breakdowns.

CHAPTER 4

**DESIGN & IMPLEMENTATION OF
PROPOSED SYSTEM**

4.1 System Overview

The proposed Nirikshak: An Industrial IoT & Predictive Maintenance Framework is continuous monitoring of machine condition, anomalies detection, faults diagnosing, and Remaining Useful Life (RUL) prediction for industrial devices. An effective predictive maintenance solution is formed by combining IIoT, embedded systems, edge computing, machine learning, cloud storage, and real-time visualization.

Continuous monitoring of machine state and early detecting of anomalies become the primary goal of the proposed framework, which is achieved in order to prevent machine failure. Traditional approaches often lead to unnecessary downtimes and high maintenance costs. In order to eliminate such disadvantages, the framework allows performing condition-based maintenance thanks to continuous data collecting and analysis.

Machine health indicators, which include temperature, vibration, RPM, voltage, and current, are collected from several types of sensors. Data Acquisition Devices are used for data collection based on STM32F103 and transmitted via Wi-Fi module (ESP32). At the edge computing level, machine learning algorithms perform fault classification, anomalies detection, and RUL prediction. All results are synchronized with Firebase Realtime Database and visualized in the form of web dashboard.

4.2 System Architecture

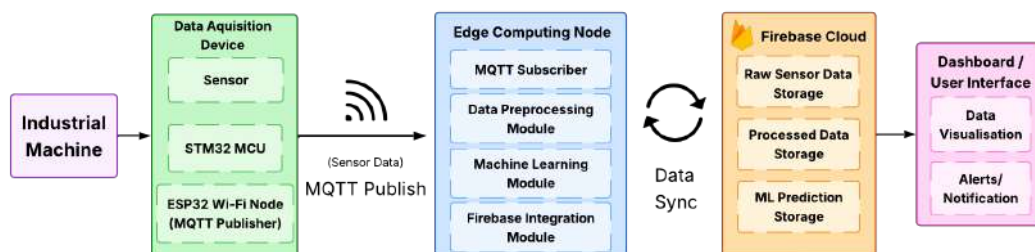


Figure 4.1: System Architecture of Nirikshak Predictive Maintenance Framework

Nirikshak architecture is designed to leverage a tiered structure of IIoT technology architecture, which contain data acquisition layer, communication layer, edge processing layer, cloud storage layer, and visualization layer. This architecture facilitates continuous monitoring of machine state and allows predictive maintenance analysis.

1. Industrial Machine

In the case of industrial machinery, the monitored equipment includes induction motors, pumps, compressors, generators, and rotational equipment. When

the machinery is operating, it generates physical and electrical variables indicating its condition. Any change in the generated variables shows any kind of deterioration or malfunctioning in the machinery.

2. Data Acquisition Device

The Data Acquisition Device acts as the hardware interface connecting the industrial machines to the predictive maintenance system. It continuously acquires machine condition parameters and packages them for transferring them to the edge computing node. The data acquisition device includes sensing modules, microcontroller unit, and wireless communication interface to ensure reliable data acquisition and communication.

- **Sensors** A number of sensors are used to collect the required indicators of machine health. The DS18B20 temperature sensor acquires changes in the machine's temperature to assist in determining when the machine may be developing overheating issues. The MPU6050 sensor collects vibrations that are helpful in identifying imbalance, misalignment, bearing failure, and mechanical deterioration. An inductive proximity sensor collects the revolutions per minute (RPM) of the machine's shaft. Electrical indicators are acquired through the use of ZMPT101B voltage sensor and SCT-013 current sensor to determine power consumption levels. In summary, these sensors collectively gives all information about the machine's operating conditions.
- **STM32F103 Microcontroller** The STM32F103 microcontroller acts as the master controller of the data acquisition device. All sensor modules are connected to the microcontroller and machine condition parameter acquisition occurs locally. The microcontroller performs real-time sensor readings, analog-to-digital conversions, calibration processes, validation of acquired data, and structuring the sensor readings into a proper format. This way, the STM32 ensures that only reliable information is transferred to the communication node.
- **ESP32 Wi-Fi Node** The ESP32 module works as the communication interface in the framework. Machine condition data acquired through the STM32 are sent wirelessly through Wi-Fi connectivity. The ESP32 pushes the collected machine condition data to MQTT topics and controls network communication between the data acquisition node and edge computing node. The networking capabilities of the module simplify

framework implementation while providing reliable data transfer. Re-connection procedures and communication management are handled by the ESP32 module.

3. Edge Computing Node

The Edge Computing Node represents the main processing layer of the framework and implements all machine learning procedures locally. The processing layer gets machine condition data from the communication network, preprocesses the acquired information, analyzes and predicts the occurrence of possible failures, and produces maintenance intelligence.

- **MQTT Subscriber** The MQTT Subscriber is connected to the MQTT broker and continuously receives real-time sensor data published by the ESP32 communication node. It subscribes to topics, consumes incoming data streams and stores received data for later analysis. The subscriber is the entry point to the edge computing layer, which guarantees reliable collection of machine condition data. Received measurements are stored and then forwarded to subsequent processing modules for predictive maintenance analysis.
- **Data Preprocessing Module** Sensor data is pre-processed prior to machine learning analysis to improve data quality and model performance. This phase extracts useful features from the incoming sensor packets, removes incomplete or invalid records, normalizes feature values, reduces noise, and parses the sensor packets. Proper preprocessing ensures consistency across datasets and allows machine learning models to make more accurate predictions. The processed data will be used as input for algorithms of predictive maintenance.
- **Machine Learning Module** Machine Learning Module is the smart component of the architecture. This module analyzes the processed data coming from the sensors in order to determine the status of the machine, as well as its future operating conditions. The engine is capable of diagnosing machine problems using fault classification techniques, detecting abnormal machine behavior via anomaly detection methods, and predicting the life span of the machine prior to failure using remaining useful life estimation techniques. Machine learning module takes the raw sensor data and transforms it into maintenance-related information.
- **Firebase Integration Module** The Firebase Integration Module is responsible for synchronizing processed sensor data and machine learning results with the cloud platform. After analysis is completed, the module formats prediction outputs, updates database records, uploads machine health information, and stores anomaly detection results. This module

ensures seamless communication between the edge computing environment and the cloud infrastructure, enabling remote accessibility and centralized data management.

4. **Firestore Cloud Platform** Firestore Realtime Database is the centralized cloud storage platform of the framework. Raw sensor measurements, processed data, machine learning predictions, anomaly information and historical operational records are stored. The cloud platform allows real-time synchronization between multiple system components and secure remote access to machine condition information. Historical records stored in the database support long-term trend analysis, maintenance planning and performance evaluation.
5. **Dashboard/User Interface** The Dashboard offers real-time visualization and monitoring for machine operators and maintenance personnel. The web-based interface pulls data from Firestore Realtime Database and displays machine condition data using interactive charts, indicators and status panels. They can be used for real-time sensor data monitoring, machine health states, fault prognostics, anomaly detection, historical performance trends and Remaining Useful Life. The dashboard enables maintenance teams to make informed decisions based on real-time operational insights and helps support proactive maintenance planning to improve equipment reliability and reduce operational expenses.

4.3 Flow Chart

The operation of the proposed Nirikshak framework follows a systematic sequence of steps for real-time machine condition monitoring and predictive maintenance, as illustrated in Figure 4.2.

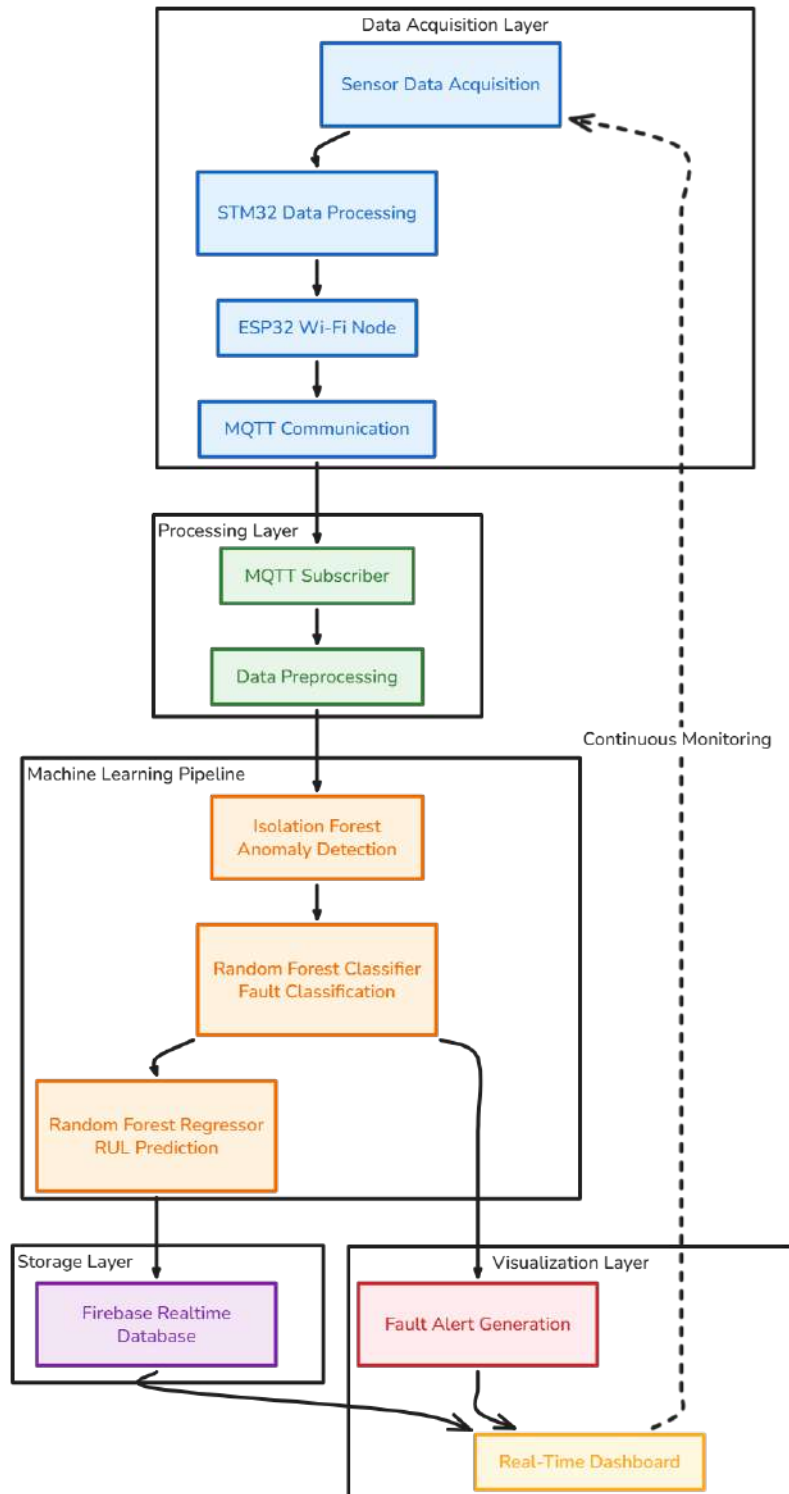


Figure 4.2: Flowchart of Nirikshak Predictive Maintenance Framework

1. The system continuously acquires machine health parameters such as temperature, vibration, current, voltage, and rotational speed (RPM) using multiple industrial sensors connected to the data acquisition unit.
2. The acquired sensor readings are processed by the STM32F103 microcontroller, where sensor interfacing, analog-to-digital conversion, data filtering, and formatting operations are performed.
3. The processed data is transmitted to the ESP32 Wi-Fi module through UART communication. The ESP32 acts as the communication interface between the embedded hardware and the monitoring system.
4. The ESP32 publishes the sensor readings to the MQTT broker, enabling reliable real-time wireless communication between the data acquisition device and the edge computing node.
5. A Python-based MQTT subscriber receives the incoming sensor data and performs data ingestion and preprocessing operations, including data validation, formatting, normalization, and preparation for machine learning analysis.
6. The preprocessed data is first analyzed using the Isolation Forest anomaly detection model to identify abnormal operating conditions and generate an anomaly score indicating the likelihood of abnormal machine behavior.
7. The processed data is then supplied to the Random Forest Classifier, which identifies the machine's operating condition and classifies potential fault types such as bearing faults, overheating conditions, or electrical abnormalities.
8. Subsequently, the Random Forest Regressor estimates the Remaining Useful Life (RUL) of the machine, providing an indication of the expected operational duration before maintenance or replacement is required.
9. The generated anomaly status, fault classification results, and RUL predictions are stored in the Firebase Realtime Database. If a fault condition is detected, the system generates an alert notification for maintenance personnel.
10. The dashboard retrieves real-time sensor data and prediction results from Firebase and displays machine health information, fault alerts, anomaly status, and predictive maintenance insights.
11. The monitoring process continues continuously, enabling real-time machine condition assessment and predictive maintenance decision support.

4.4 Electronic Components Used

The Nirikshak framework consists of sensors, microcontrollers, communication modules, and power supply units that work together to monitor machine health in real time. These components enable reliable data acquisition, processing, and transmission for predictive maintenance applications.

1. STM32F103 Development Board

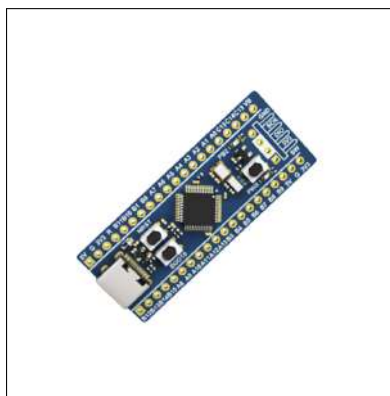


Figure 4.3: STM32F103 Development Board

STM32F103C8T6 is a 32-bit ARM Cortex-M3 microcontroller operating at 72 MHz. It acts as the main controller of the system by acquiring sensor data, processing it, and transmitting it to the communication module for further analysis.

2. MPU6050 Vibration Sensor



Figure 4.4: MPU6050 Vibration Sensor

MPU6050 is a six-axis MEMS sensor that combines a three-axis accelerometer and a three-axis gyroscope. It is used to measure machine vibration and motion, enabling continuous condition monitoring and fault detection.

3. ESP32 WiFi Node



Figure 4.5: ESP32 Development Board

The ESP32 is a Wi-Fi and Bluetooth-enabled microcontroller used as the communication interface in the proposed framework. It receives sensor data from the STM32F103 via UART and transmits it wirelessly using the MQTT protocol. Its networking capabilities enable real-time data transfer, remote monitoring, and Industrial IoT connectivity between the data acquisition unit and the edge computing node.

4. DS18B20 Temperature Sensor



Figure 4.6: DS18B20 Temperature Sensor

DS18B20 is a digital temperature sensor with one-wire communication capability. The sensor converts temperature into digital data and transmits it directly to the microcontroller. It is used to monitor machine temperature and identify overheating conditions that may affect equipment performance and reliability.

5. Inductive Proximity Sensor (RPM Sensor)



Figure 4.7: Inductive Proximity Sensor (RPM Sensor)

An inductive proximity sensor detects metallic objects without physical contact. It generates pulses whenever a rotating metallic object passes near the sensing surface. The sensor is used for rotational speed (RPM) measurement by counting the generated pulses and determining the speed of rotating machinery.

6. Voltage Sensor Module (ZMPT101B)

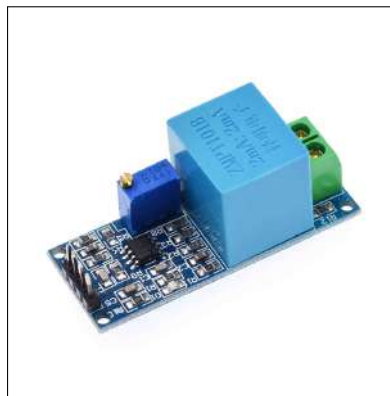


Figure 4.8: Voltage Sensor Module

The ZMPT101B voltage sensor module measures AC voltage supplied to industrial equipment. The input voltage is scaled down to a safe range suitable for microcontroller ADC inputs. It is used for voltage monitoring and power quality assessment by continuously measuring supply voltage and detecting abnormal voltage conditions.

7. SCT-013 Current Sensor



Figure 4.9: SCT-013 Current Sensor

SCT-013 is a non-invasive current transformer sensor designed to measure alternating current without direct electrical contact. The sensor generates a proportional output signal corresponding to the current flowing through the conductor. It is used for monitoring machine current consumption and helps in assessing equipment load conditions and electrical performance.

8. ST-Link Programmer



Figure 4.10: ST-Link Programmer

ST-Link is a programming and debugging tool designed for STM32 microcontrollers. It communicates with the microcontroller through the Serial Wire Debug (SWD) interface and uploads firmware to the target device. The tool is used for program development, debugging, testing, and firmware updates, enabling efficient software deployment and troubleshooting during system development.

9. Experimental Test Motor



Figure 4.11: Single Phase AC Motor

A single-phase AC fan motor was used as the test machine for demonstrating the proposed predictive maintenance framework. The motor serves as the industrial equipment under monitoring and provides real-time operating parameters such as temperature, vibration, current consumption, supply voltage, and rotational speed (RPM). These parameters are continuously acquired through the data acquisition system and analyzed using machine learning algorithms to assess machine health and predict potential faults.

The selected motor is a high-speed all-purpose fan motor rated at 230 VAC and 100 W. The motor is equipped with a 4 μ F capacitor for starting and running operation. Due to its continuous rotational operation and ease of experimentation, it provides a suitable platform for validating the proposed Industrial IoT-based predictive maintenance framework.

Table 4.1: Experimental Test Motor Specifications

Parameter	Specification
Motor Type	Single-Phase AC Fan Motor
Rated Voltage	230 VAC
Rated Power	100 W
Supply Frequency	50 Hz
Capacitor Rating	4 μ F, 450 VAC
Mounting Type	Flange Mount
Cooling Method	Self-Air Cooled
Application	Predictive Maintenance Demonstration

4.5 PCB Design and Development

A custom Printed Circuit Board (PCB) was designed to integrate the STM32F103 microcontroller, ESP32 communication module, sensor interfaces, power supply circuitry, and signal conditioning components into a single compact hardware platform. The PCB serves as the Data Acquisition (DAQ) device of the Nirikshak framework and simplifies sensor interfacing while improving system reliability and maintainability.

4.5.1 PCB Schematic Design

The schematic was developed using EasyEDA based on the hardware requirements of the predictive maintenance framework. The STM32F103 Black Pill microcontroller acts as the central controller and interfaces with all sensing modules. The ESP32 module provides wireless communication through the MQTT protocol. Dedicated interfaces were incorporated for vibration, temperature, voltage, current, and RPM sensors.

The schematic also includes:

- STM32F103 Black Pill microcontroller
- ESP32 DevKit communication module
- DS18B20 temperature sensor interface
- MPU6050 vibration sensor interface
- ZMPT101B voltage sensor interface
- SCT-013 current sensor conditioning circuit
- Inductive proximity sensor interface for RPM measurement
- Power supply and switching circuitry

4.5.2 PCB Layout Design

After schematic verification, the PCB layout was generated in EasyEDA. Component placement was optimized to minimize routing complexity and facilitate sensor connectivity. The STM32F103 controller was positioned at the center of the board, while sensor connectors and power interfaces were placed along the edges for easy access. Separate routing paths were maintained for sensor signals, power distribution, and communication lines to improve overall system organization.

The final PCB layout integrates all required subsystems into a compact DAQ device suitable for industrial monitoring applications.

4.5.3 Advantages of the Custom PCB

The developed PCB provides several advantages:

- Compact integration of all hardware components
- Reduced wiring complexity
- Improved reliability and maintainability
- Easy sensor interfacing and replacement
- Simplified deployment in industrial environments
- Dedicated communication and power management circuitry

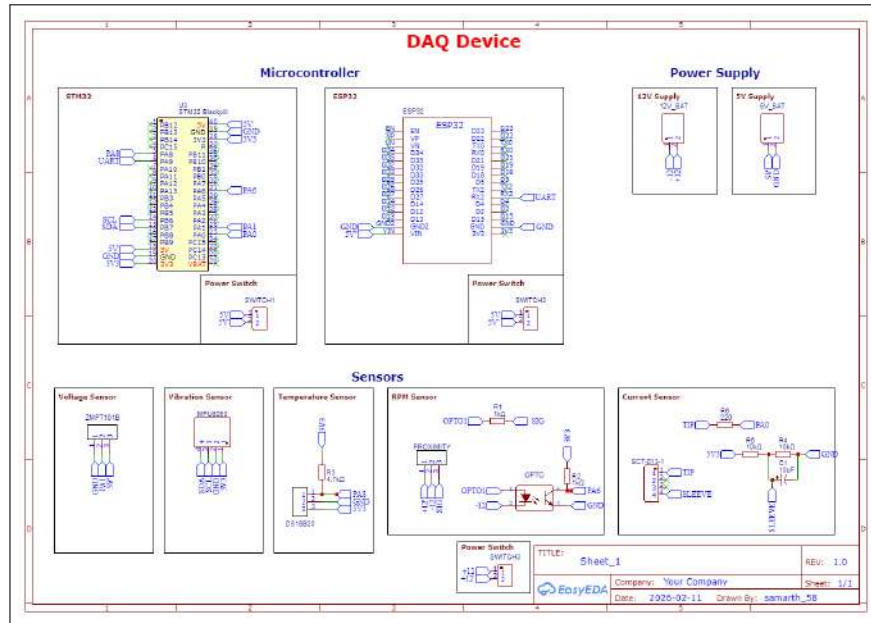


Figure 4.12: DAQ Device Schematic

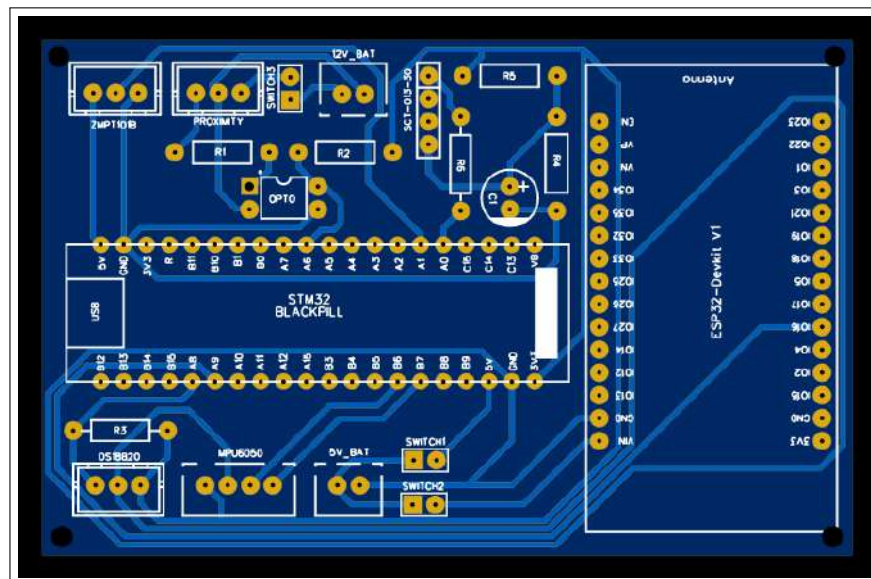


Figure 4.13: DAQ Device PCB Layout

CHAPTER 5

SOFTWARE IMPLEMENTATION

5.1 Introduction

The Framework acts as the intelligence layer of the entire system. It is the part of the system that collects the data from Data Acquisition Device, processes the collected data, executes the machine learning algorithm, maintains the operational logs and displays the live machine health information using a web dashboard. The software design uses a hybrid edge-cloud model where the sensor data obtained from the industrial machines will be transferred through MQTT protocol and collected by a processor engine made up of Python programming language. The collected data is then pre-processed for cleaning, normalization, outlier removal and feature engineering before being sent to the machine learning model. The predictions generated by the machine learning models are then saved in the Firebase Realtime Database and displayed through the real-time monitoring dashboard.

5.2 Software Required

Table 5.1 presents the software tools utilized during the development and implementation of the Nirikshak framework.

Table 5.1: Software Requirements

Sr. No.	Software	Purpose
1	STM32CubeIDE	Firmware development for STM32F103
2	Arduino IDE	ESP32 programming and debugging
3	Python 3.10	Data processing and machine learning
4	Visual Studio Code	Application development environment
5	Flask Framework	Backend web server implementation
6	Flask-SocketIO	Real-time dashboard communication
7	Firebase Realtime Database	Cloud data storage
8	Mosquitto MQTT Broker	Message communication between devices
9	Scikit-Learn	Machine learning model development
10	Pandas	Data manipulation and preprocessing
11	NumPy	Numerical computation
12	Matplotlib	Data visualization and analysis
13	Git	Version control
14	GitHub	Project repository management
15	HTML5, CSS3, JavaScript	Dashboard development

5.3 Software Specification

The software specifications used for implementation are presented in Table 5.2.

Table 5.2: Software Specifications

Parameter	Specification
Operating System	Windows 11 / Ubuntu 24.04 LTS
Programming Language	Python 3.10
Backend Framework	Flask 2.0+
Real-Time Communication	Flask-SocketIO
MQTT Protocol	MQTT v3.1.1
MQTT Broker	Mosquitto MQTT Broker
Database	Firebase Realtime Database
Frontend Technologies	HTML5, CSS3, JavaScript
Machine Learning Library	Scikit-Learn
Data Processing Library	Pandas
Numerical Computing	NumPy
Version Control	Git
Development Environment	Visual Studio Code
Data Format	JSON, CSV
Communication Security	TLS Encryption
Supported Browser	Chrome, Edge, Firefox

5.4 Software Implementation

The software implementation consists of five major modules: MQTT Communication Module, Data Preprocessing Module, Machine Learning Module, Cloud Storage Module, and Dashboard Visualization Module.

5.4.1 MQTT Communication Module

The MQTT Communication Module enables wireless communication between the Data Acquisition Device and the Edge Computing Node. The ESP32 Wi-Fi node transmits sensor data for temperature, vibration, voltage, current, and RPM values to the MQTT broker. The Python subscriber listens continuously to the predefined MQTT topic and gets real-time data of machine condition.

The subscriber receives sensor readings, validates the incoming payload and converts it into a structured JSON format for further processing. The data received is simultaneously stored in a local CSV file for backup storage in case of network interruptions.

5.4.2 Data Preprocessing Module

Data Preprocessing Module The Data Preprocessing Module prepares the raw sensor data for the machine learning analysis. First, missing values are identified and filled using median imputation techniques from historical operational data. Then the sensor parameters are normalized to a common schema, to ensure all data records are consistent.

Detection of abnormal operating conditions is performed based on rules for detecting outliers. The Temperature, Vibration, Voltage, Current and RPM values are threshold based validation. Once validated, we perform the feature scaling process using the StandardScaler algorithm, to normalize all the sensor values in order to improve the machine learning performance.

The module also calculates the machine operating hours by keeping track of timestamps and cumulative runtime information which is then used for Remaining Useful Life prediction.

5.4.3 Machine Learning Module

The Machine Learning Module acts as the brain of the Nirikshak architecture. This module helps in anomaly detection, classification of machine failure, and predicting the Remaining Useful Life (RUL) of industrial machines by using the processed information from sensors. A three-step machine learning process has been proposed for this purpose.

1. Isolation Forest Based Anomaly Detection

The first phase of the predictive maintenance process uses the Isolation Forest technique for the purpose of anomaly detection. Isolation Forest is an unsupervised machine learning algorithm used to detect anomalies by constructing random decision trees.

The algorithm was selected because:

- It does not require labelled fault data.
- It efficiently detects unknown abnormal conditions.
- It provides fast inference suitable for real-time applications.
- It performs well on multidimensional sensor datasets.

The model analyzes incoming sensor readings and generates anomaly scores indicating the likelihood of abnormal machine behaviour.

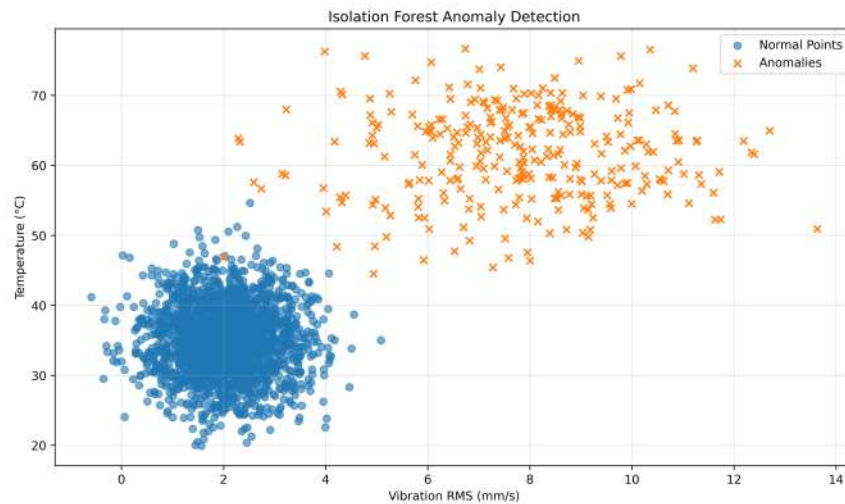


Figure 5.1: Isolation Forest Anomaly Detection Results

Table 5.3: Isolation Forest Performance

Metric	Value
Accuracy	86.58%
Precision	1.0000
Recall	0.3735
F1 Score	0.5438

The model achieved perfect precision, indicating that all detected anomalies corresponded to actual abnormal conditions. This minimizes false alarms and improves maintenance decision reliability.

2. Random Forest Based Fault Classification

Following the process of anomaly detection, the sensor data is fed into the Random Forest classifier for performing the diagnosis of faults. The Random Forest classifier is an ensemble machine learning technique based on decision trees.

The classifier identifies the following machine conditions:

- Bearing Fault
- Electrical Fault
- Misalignment
- Rotor Fault
- Overheating
- Unbalance
- Healthy Condition

The algorithm was selected because of its:

- High classification accuracy
- Resistance to overfitting
- Ability to handle nonlinear relationships
- Robustness against noisy industrial data

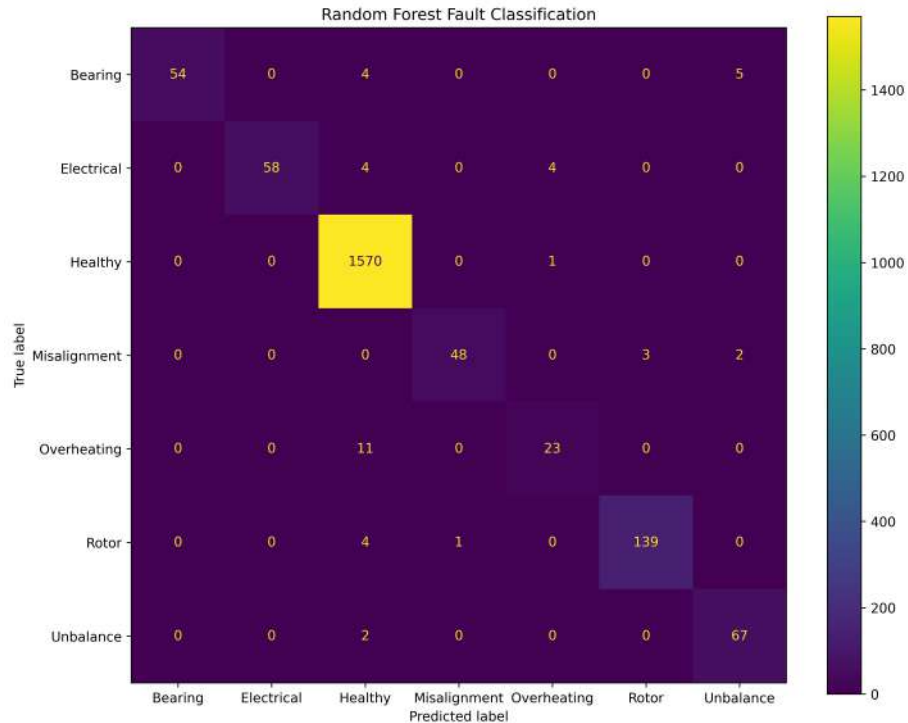


Figure 5.2: Random Forest Fault Classification Confusion Matrix

Table 5.4: Random Forest Classification Performance

Metric	Value
Accuracy	97.95%
Weighted F1 Score	0.9918

The classifier achieved excellent fault recognition capability with an overall classification accuracy of 97.95%. Most fault categories were correctly identified with minimal class confusion.

3. Random Forest Based Remaining Useful Life Estimation

The final stage of the machine learning pipeline estimates the Remaining Useful Life (RUL) of the machine. A Random Forest Regressor was employed to model degradation trends and predict future machine health.

The regressor estimates the operational duration remaining before maintenance intervention or component replacement becomes necessary.

Advantages of the selected regressor include:

- Strong nonlinear modelling capability
- Robustness against sensor noise
- Reduced overfitting
- High prediction accuracy

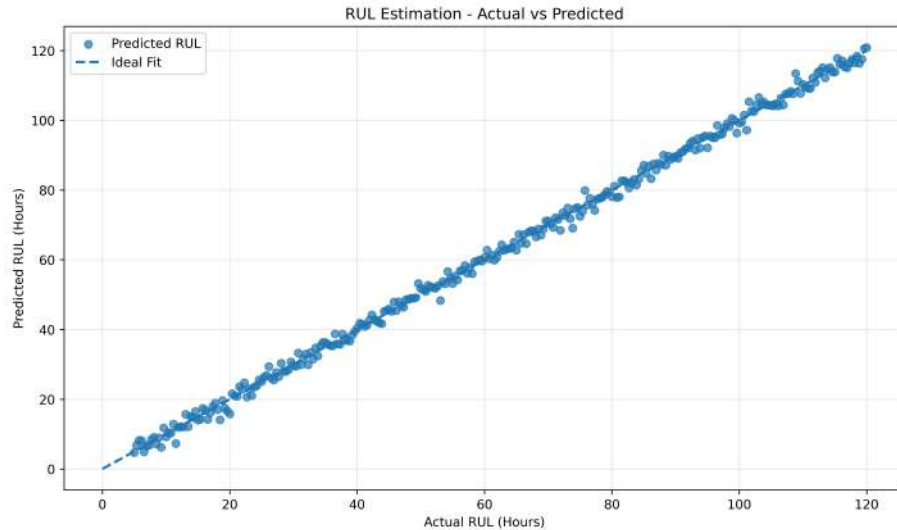


Figure 5.3: Actual vs Predicted Remaining Useful Life

Table 5.5: RUL Prediction Performance

Metric	Value
RMSE	4.52 Hours
R^2 Score	0.9973

The obtained R^2 score of 0.9973 indicates an excellent correlation between actual and predicted Remaining Useful Life values. The low RMSE value demonstrates accurate lifespan estimation suitable for predictive maintenance applications.

4. Significance of the Proposed Machine Learning Pipeline

The proposed three-stage machine learning pipeline provides a comprehensive predictive maintenance framework by integrating anomaly detection, fault classification, and Remaining Useful Life estimation into a single workflow.

The significance of the proposed approach can be summarized as follows:

- Early detection of abnormal machine behaviour.
- Accurate identification of machine fault categories.
- Reliable estimation of machine lifespan.
- Reduction of unplanned downtime.
- Improved maintenance scheduling.

- Enhanced machine reliability and operational efficiency.
- Support for Industry 4.0 based predictive maintenance systems.

The obtained results validate the effectiveness of the Nirikshak framework and demonstrate its suitability for real-time industrial machine monitoring and predictive maintenance applications.

5.4.4 Cloud Storage Module

The Cloud Storage Module synchronizes the local processing system with the cloud infrastructure. The sensor readings, anomaly detection results, fault classification outputs and Remaining Useful Life predictions are uploaded to Firebase Realtime Database.

The database is separated into raw sensor data, processed data and machine learning predictions. This segmentation improves data organization and allows the dashboard application to retrieve efficiently.

The cloud storage layer is also accessible remotely, allowing maintenance personnel to monitor machine health from anywhere there is an Internet connection.

5.4.5 Dashboard Visualization Module

The Dashboard Visualization Module provides a graphical user interface for real-time monitoring of machine conditions. The dashboard is implemented using HTML5, CSS3 and JavaScript and is connected to the backend through REST APIs and Web-Socket communication.

Dynamic representation of the sensor value readings is done using graphical charts and gauges. Real time data of the parameters of the machines, anomaly status, fault prediction and RUL is displayed on the dashboard without the necessity of refreshing the page. Graphs may also be used for displaying historical data of the sensors, enabling the operators to analyze the machine performance over an extensive period of time. The dashboard acts as the primary interface for decision making for maintenance engineers and plant operators.

5.5 Chapter Summary

This chapter described the implemented software of the Nirikshak Industrial IoT-Based Predictive Maintenance Framework. It discussed the software tools, programming environment, system specifications and implementation methodology utilized for implementation of the project. Detailed description regarding operation of the MQTT communications layer, data pretreatment pipeline, machine learning models,

cloud database integration and dashboard visualisation was provided. The software architecture being implemented will provide real-time monitoring of machines, detect anomalies, classify faults and predict Remaining Useful Life, providing an intelligent predictive maintenance solution for use in industrial environments.

CHAPTER 6

TESTING & TROUBLESHOOTING

6.1 Introduction

Testing and troubleshooting have been used to validate the performance and reliability of the proposed Nirikshak predictive maintenance framework. Testing is done with the intention of ensuring that the hardware and software elements of the developed framework function as expected and interact with each other without any problem. The developed framework includes several sensors, STM32 and ESP32 microcontrollers, MQTT communication protocol, Firebase cloud storage, machine learning models and web-based dashboard. Hence, testing became necessary to ascertain that each of the sub-components functions properly.

Testing included assessing the accuracy of sensors, effectiveness of communication, synchronization of cloud and the performance of the web-based dashboard among others. Troubleshooting of both hardware and software problems took place throughout the implementation stage. Successful testing indicated that the framework could be used for real time machine condition monitoring and predictive maintenance applications.

6.2 Testing Methodology

The testing approach for evaluating both modules individually and the entire system was systematic. First, each sensor was tested individually for the purpose of validating its capacity to capture the corresponding machine parameter. Temperature, vibration, voltage, current, and RPM sensors were validated through the comparison of their output values with the expected values at various operational states of the machine.

Following the process of validating the sensors, communication tests were done between the STM32 microcontroller and ESP32 Wi-Fi module. The collected sensor data was then sent using the MQTT protocol. Consistency and reliability of the transmission of the sensor data were checked. Also, the subscriber application on the edge computing node was tested for successful collection and storing of the sensor data. The machine learning algorithms were tested by collecting and using historical data.

The Isolation Forest algorithm was used to test for anomalies, whereas the Random Forest Classifier algorithm was used to test for faults. The final test carried out was that of end-to-end tests.

6.3 Testing Procedure

In order to test the system, all sensors had to be connected to the STM32F103 microcontroller and powered on. The industrial machine was run normally, and the readings from the sensors were monitored throughout the whole process. The data collected was sent to the ESP32 module and then to the MQTT broker.

The data received by the MQTT subscriber application was sent to the Firebase Realtime Database for further processing. The machine learning models were processing the input data and making predictions related to the machine health status. The results of the prediction were then uploaded to the database and shown on the monitoring dashboard.

In order to test the efficiency of the system, abnormal operating conditions like elevated levels of vibrations and temperatures had to be simulated. The input data produced was analyzed by the machine learning models and classified for anomalies and possible faults. The dashboard was monitored to make sure that the results are shown correctly.

6.4 Troubleshooting

Some of the problems that arose in the process of implementation and testing include noisy and fluctuating values from the sensors caused by environmental factors and interference from electrical sources. Calibration and filtering through software were used to solve this issue and improve the accuracy of measurements.

Problems related to the loss of WiFi connection resulting in the disconnection of MQTT were solved by ensuring the automatic reconnection capability of the ESP32 software. Delay in data synchronization between the firebase and the dashboard was also experienced while sending the data continuously. Improved synchronization was achieved by optimizing the database updating interval.

Inefficient machine learning algorithms were due to the presence of noise from the sensors. Noise reduction and normalization were conducted on the collected data to improve the efficiency of the machine learning models.

6.5 Results & Analysis

The developed Nirikshak framework was successfully tested under various operating conditions to evaluate its sensing, communication, cloud integration, dashboard visualization, and machine learning capabilities. The obtained results demonstrate the effectiveness of the proposed Industrial IoT-based predictive maintenance system for real-time machine condition monitoring.

6.5.1 Sensor Data Analysis

Parameters such as temperature, vibration, RPM, current, and voltage of machine were continuously monitored while the system was operational. The sensor data captured was successfully transferred through the MQTT network to Firebase Real-time Database.

1. Temperature Analysis

Figure 6.1 shows the temperature variation recorded during machine operation. During normal operating conditions temperature remained stable and increased during simulated overheating scenarios. The observed trend demonstrates the capability of the DS18B20 sensor to monitor thermal behaviour effectively.

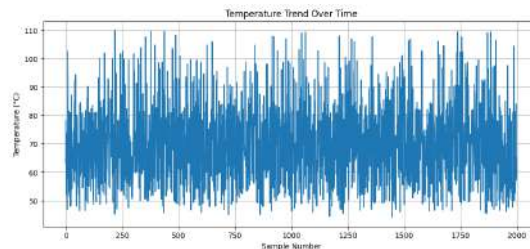


Figure 6.1: Temperature Trend Analysis

2. Vibration Analysis

Vibration is one of the most important indicators of machine health. Figure 6.2 illustrates the vibration pattern measured using the MPU6050 sensor. Increased vibration amplitudes were observed during abnormal operating conditions, indicating potential imbalance and mechanical faults.

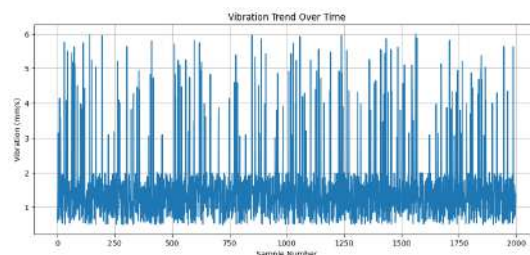


Figure 6.2: Vibration Trend Analysis

3. RPM Analysis

The rotational speed of the machine was measured using the inductive proximity sensor. Figure 6.3 shows the RPM variation observed during testing. Minor fluctuations were recorded under varying load conditions, while significant deviations were observed during simulated fault scenarios.

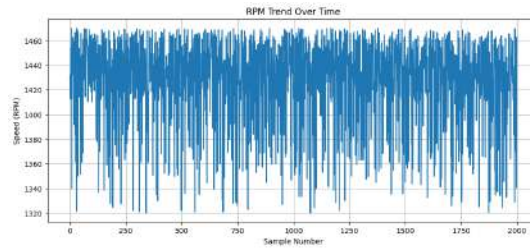


Figure 6.3: RPM Trend Analysis

4. Current Analysis

Current consumption was monitored using the SCT-013 current sensor. Figure 6.4 presents the current variation during machine operation. Increased current values were observed under overload conditions, indicating higher power consumption and increased mechanical stress.

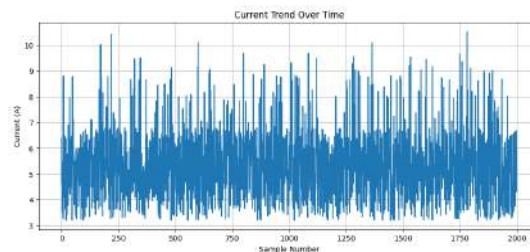


Figure 6.4: Current Trend Analysis

5. Voltage Analysis

The ZMPT101B voltage sensor continuously monitored the machine supply voltage. Figure 6.5 shows the voltage profile obtained during testing. The voltage remained relatively stable with minor fluctuations, confirming reliable power supply conditions.

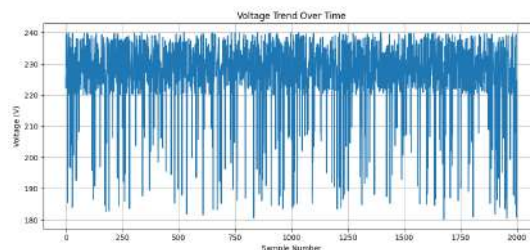


Figure 6.5: Voltage Trend Analysis

6.5.2 Dashboard Validation

The web-based dashboard successfully displayed real-time machine condition information retrieved from firebase realtime database. Sensor readings, machine health

indicators, fault predictions, anomaly status, and remaining useful life estimates were updated dynamically without requiring page refresh.



Figure 6.6: Real-Time Monitoring Dashboard



Figure 6.7: Real-Time Monitoring Dashboard

The dashboard provided a centralized interface for monitoring machine performance and enabled operators to visualize both live and historical machine condition data.

6.5.3 Machine Learning Performance Summary

The machine learning models were tested using historical as well as real-time data collected from the sensors. The obtained performance metrics indicate that the proposed predictive maintenance framework effectively analyzes machine health and predicts potential faults. This enables proactive maintenance planning, helping industries reduce downtime, lower maintenance costs, and enhance equipment reliability.

Table 6.1: Performance Evaluation of Machine Learning Models

Machine Learning Model	Application	Performance
Isolation Forest	Anomaly Detection and Outlier Identification	86.58% Accuracy
Random Forest Classifier	Fault Detection and Classification	97.95% Accuracy
Random Forest Regressor	Remaining Useful Life (RUL) Prediction	$R^2 = 0.9973$

The anomaly detection model successfully identified abnormal operating conditions while minimizing false alarms. The Random Forest Classifier accurately distinguished between multiple fault categories and achieved excellent classification performance. The Random Forest Regressor provided highly accurate Remaining Useful Life predictions, demonstrating strong predictive capability.

6.5.4 Overall System Performance

Below table summarizes the performance of the complete predictive maintenance framework.

Table 6.2: Overall System Performance Evaluation

Parameter	Result
Temperature Monitoring	Successful
Vibration Monitoring	Successful
RPM Monitoring	Successful
Current Monitoring	Successful
Voltage Monitoring	Successful
MQTT Communication	Successful
Firebase Synchronization	Successful
Dashboard Visualization	Successful
Anomaly Detection Accuracy	86.58%
Fault Classification Accuracy	97.95%
RUL Prediction (R^2 Score)	0.9973

The experimental results validate the effectiveness of the proposed Nirikshak framework. The integrated system successfully performed real-time machine monitoring, anomaly detection, fault classification, cloud synchronization, dashboard visualization, and Remaining Useful Life estimation. The obtained results demonstrate

the suitability of the framework for Industrial IoT-based predictive maintenance applications.

CHAPTER 7

ADVANTAGES & APPLICATIONS

Predictive maintenance has become an essential component of modern industrial systems as it helps organizations minimize unexpected equipment failures and improve operational efficiency. The proposed Nirikshak: An Industrial IoT & Predictive Maintenance Framework integrates Industrial IoT, edge computing, cloud connectivity, and machine learning to provide continuous monitoring and intelligent fault prediction. By analyzing real-time machine parameters, the system enables timely maintenance actions, reduces downtime, and improves equipment reliability. Its low-cost and scalable architecture makes it suitable for both small and large industrial environments.

7.1 Advantages

The advantages of the proposed system can be summarized as follows:

fault detection and prediction, enabling timely maintenance before critical failures occur.

1. Reduction in machine downtime through proactive maintenance scheduling and condition-based monitoring.
2. Lower maintenance and operational costs by avoiding unnecessary preventive maintenance and costly breakdowns.
3. Real-time monitoring of machine health using multiple sensors for continuous observation of operating parameters.
4. Remote monitoring through cloud connectivity, allowing operators to access machine status from any location.
5. Improved reliability and equipment performance by continuously tracking machine conditions and degradation patterns.
6. Scalable and flexible system architecture that can be expanded to support multiple machines and industrial environments.
7. Cost-effective implementation using low-cost hardware and open-source technologies, making it suitable for small and medium enterprises.
8. AI-based fault diagnosis and predictive analytics using machine learning algorithms for anomaly detection, fault classification, and Remaining Useful Life (RUL) estimation.
9. Industry 4.0 and Industrial IoT compatibility, supporting smart manufacturing and digital transformation initiatives.

7.2 Applications

The proposed system can be employed in a variety of applications such as:

1. Manufacturing industries for monitoring production equipment and reducing downtime.
2. Automotive production facilities for predictive maintenance of assembly line machinery and industrial robots.
3. Power generation plants for monitoring generators, turbines, motors, and auxiliary equipment.
4. Process and chemical industries for ensuring reliable operation of pumps, compressors, and processing units.
5. Smart factories and Industry 4.0 environment for intelligent asset management and automated maintenance planning.
6. HVAC and industrial cooling systems for monitoring temperature, vibration, and equipment performance.
7. Water treatment and pumping stations for continuous supervision of pumps, motors, and control systems.
8. Renewable energy systems such as wind turbines and solar power plants for equipment health assessment.
9. Warehouse automation and material handling systems for monitoring conveyors, motors, and automated machinery.
10. Educational and research laboratories for experimentation, training, and development of Industrial IoT and predictive maintenance technologies.

CHAPTER 8

CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The project Nirikshak: An Industrial IoT Predictive Maintenance Framework has been developed and successfully implemented as an integrated solution for real-time monitoring of machine conditions and predictive maintenance. In the designed framework, Industrial IoT, edge computing, cloud computing, and machine learning are used to monitor important machine parameters and detect any possible faults before failure happens. The developed system provides continuous data collection, analysis, and monitoring. Thus, the unplanned downtime is reduced and the operational efficiency is increased. It is proved that the low-cost predictive maintenance solution for Industry 4.0 can be implemented in industry and SMEs

The major outcomes of the project are summarized as follows:

1. Successfully developed an Industrial IoT-based predictive maintenance framework.
2. Implemented real-time monitoring of temperature, vibration, RPM, voltage, and current parameters.
3. Integrated STM32F103 and ESP32 modules for efficient data acquisition and wireless communication.
4. Established reliable MQTT-based communication for real-time sensor data transmission.
5. Developed an edge computing module for data preprocessing, feature extraction, and analysis.
6. Implemented machine learning models for fault detection and predictive maintenance applications.
7. Enabled anomaly detection and Remaining Useful Life (RUL) estimation for machine health assessment.
8. Integrated Firebase cloud platform for remote data storage, synchronization, and accessibility.
9. Developed a dashboard for real-time visualization, monitoring, and maintenance decision support.
10. Demonstrated a cost-effective, scalable, and Industry 4.0-compatible solution for predictive maintenance.

8.2 Future Scope

The proposed system provides a strong foundation for intelligent predictive maintenance; however, several enhancements can be incorporated to further improve its functionality and industrial applicability.

1. Deployment of the framework in large-scale industrial environments for validation under real operating conditions.
2. Integration of additional industrial sensors such as pressure, humidity, acoustic, and thermal imaging sensors for comprehensive equipment monitoring.
3. Development of a dedicated mobile application for remote monitoring, alerts, and maintenance management.
4. Implementation of advanced deep learning models such as CNN, LSTM, Transformer, and hybrid architectures for improved prediction accuracy.
5. Integration of digital twin technology to create virtual representations of industrial assets and enable advanced predictive analytics.
6. Extension of the framework to support multi-machine and multi-site monitoring within a unified platform.
7. Optimization of Edge AI algorithms for faster inference, reduced latency, and lower computational requirements.
8. Integration with enterprise-level systems such as ERP, MES, and SCADA platforms for automated maintenance planning and industrial workflow management.

The future enhancements will further improve the intelligence, scalability, and reliability of the Nirikshak framework, enabling broader adoption of predictive maintenance technologies across modern smart manufacturing environments.

CHAPTER 9

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ANNEXURE A
PLAGIARISM REPORT



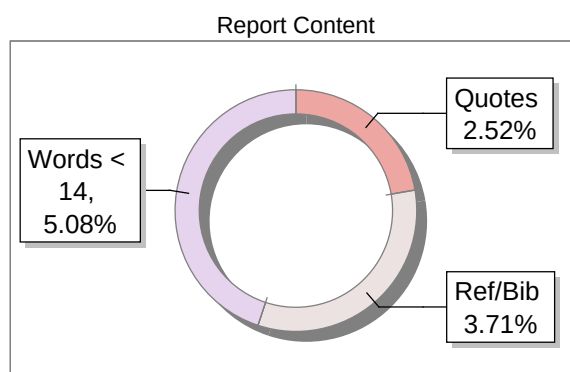
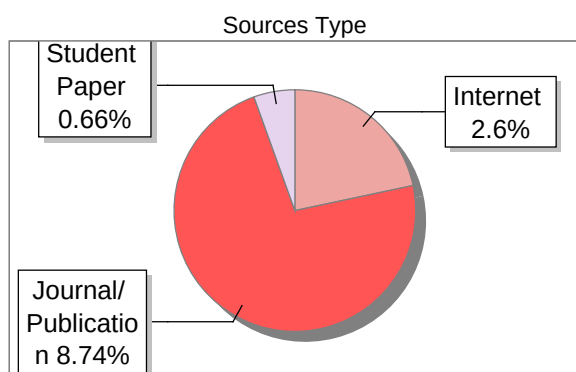
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Description of the Work:

To develop a lowcost, customizable Industrial IoT system that collects realtime machine data, predicts failures using machine learning, and displays key parameters through a userfriendly dashboard.

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NIRIKSHAK: AN INDUSTRIAL IOT AND PREDICTIVE MAINTENANCE FRAMEWORK

ABSTRACT:

The fast advancement of the Industrial Internet of Things (IIoT) has allowed industries to transition from reactive and preventive maintenance approaches to predictive maintenance (PdM) systems that guarantee enhanced efficiency, minimized downtime, and streamlined operational expenses. This initiative suggests a predictive maintenance framework based on IIoT, which combines data acquisition (DAQ) systems involving microcontrollers and sensors with wireless communication for monitoring in real-time. Machine metrics like vibration, temperature, and current are continuously gathered and sent to an edge computer that runs a cloud platform for data analysis. Algorithms in machine learning are utilized to examine these parameters, identify anomalies, and forecast possible failures prior to their occurrence. The outcomes are shown on an engaging dashboard that offers visualization, trend analysis, and notifications for maintenance planning. The system guarantees ongoing oversight, immediate decision-making, and anticipatory fault prevention. This system seeks to improve industrial efficiency, reduce unexpected failures, and promote data-centered maintenance approaches in line with Industry 4.0 goals

1. Introduction:

The rise of the Industrial Internet of Things (IIoT) has transformed the operational environment of industries, marking the dawn of Industry 4.0. By facilitating connectivity among machines, sensors, controllers, and cloud platforms, IIoT offers immediate insight into industrial assets and processes. Among its numerous applications, predictive maintenance (PdM) has garnered considerable interest as it directly affects the productivity, reliability, and cost efficiency of operations.

Historically, industries depended on corrective maintenance (repairing a machine post-failure) or preventive maintenance (planning repairs at regular intervals irrespective of condition). Corrective maintenance causes expensive downtimes, whereas preventive maintenance can lead to needless servicing and resource wastage. Predictive maintenance utilizes sensor-based data analysis to assess the remaining useful lifespan of machinery, identify initial symptoms of deterioration, and anticipate potential malfunctions. This enables maintenance teams to act solely when needed, achieving a balance between operational efficiency and cost reduction.

The crucial factor for predictive maintenance is the collection and analysis of real-time data. Sensors integrated into equipment constantly track factors like vibration, temperature, current, and pressure. These signals are obtained through microcontroller-driven data acquisition (DAQ) systems and sent wirelessly to cloud or edge platforms. Sophisticated machine learning techniques examine the data to detect unusual patterns, trends, and irregularities. The findings are showcased via interactive dashboards, offering engineers actionable insights and recommendations.



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PdM frameworks based on IIoT have gained significant importance in sectors where downtime results in considerable losses, including manufacturing, energy, and transportation. Through the integration of real-time tracking, wireless connectivity, and AI-powered analysis, PdM minimizes unexpected breakdowns, prolongs the lifespan of equipment, and improves safety in the workplace. Nonetheless, issues like dependable connectivity, precise model training, and compatibility with existing industrial systems need to be tackled for effective implementation.

This initiative seeks to create an affordable IIoT and predictive maintenance system that functions continuously, offering industries immediate machine health data, failure forecasts, and automated decision-making assistance to enhance efficiency and robustness.

2. Literature Survey:

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		Paper Title	Methodology	Scope for Improvement
1.	Soumya Naiya (2024)	AI-Powered Predictive Maintenance in IoT-Enabled Smart Factories	Discusses AI and IoT integration for real-time predictive maintenance in smart factories; uses ML models like CNN, RNN; shares case studies	Improve data privacy & security; reduce setup costs; address skill shortages; enhance scalable edge computing and federated learning
2.	Sasmita Pani, Omkar Pattnaik, Binod Pattanayak. (2024)	Predictive Maintenance in Industrial IoT Using Machine Learning Approach	Focuses on machine learning methods for predictive maintenance using sensor data; covers data collection, preprocessing, model training, and maintenance recommendation.	Enhance data quality and sensor reliability; manage high costs; improve integration complexity; continuously update models for accuracy
3.	Hector, Ida, Rukmani Panjanathan. (2024)	Predictive Maintenance in Industry 4.0: A Survey of Planning Models and Machine Learning	Reviewed planning models for predictive maintenance. Covered how IoT devices collect data and how ML methods (supervised, unsupervised, deep learning) are used for fault detection, anomaly detection, and remaining useful life prediction.	Models need to be faster and lighter for real-time use. Better integration of IoT with ML/DL. Reduce data needs and cost. Create practical PdM strategies for industries.



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4.	Nandini Nalawade, Swati D. Jakkan, Shweta Mangnale, Pradnya Patil. (2020)	Predictive Maintenance for Industrial Equipment using IIoT, AI and ML.	Applies machine learning (Random Forest, LSTM, SVM) and IoT sensors to predict and prevent failures; evaluates models and real-time monitoring	Tackle data quality issues; improve model scalability; reduce complexity; enhance interpretability and industry adaptability
5.	Nangia, Shikhil, Sandhya Makkar, and Rohail Hassan. (2020)	IoT based Predictive Maintenance in Manufacturing Sector	Collected sensor data (temperature, vibration, etc.) from machines; applied ML models (SVM, trees) to predict failures; deployed predictive model in manufacturing.	Scale model across multiple assets; include prescriptive maintenance actions; use big data analytics for scalability
6.	Him, Leong Chee, Yu Yong Poh, and Lee Wah Pheng. (2019)	IoT-based Predictive Maintenance for Smart Manufacturing	Deployed battery-powered IIoT sensor nodes in power plant; used IPv6-based protocols for data collection; monitored latency and battery life; enabled plug & play.	Extend to other manufacturing equipment; add more diverse sensors; automate labeling to improve prediction accuracy

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The reviewed studies demonstrate that IoT-enabled predictive maintenance combined with machine learning and deep learning models (SVM, Random Forest, LSTM, CNN, RNN, etc.) significantly improves fault prediction, anomaly detection, and real-time monitoring. Most works focus on sensor-based data collection and algorithmic evaluation, while surveys provide broader insights into planning models and Industry 4.0 applications. Despite these advancements, there remains a gap in developing a holistic and scalable PdM framework that integrates data acquisition, processing, and visualization-which forms the basis of the proposed project.

3. Problem statement:

To develop a low-cost, customizable Industrial IoT system that collects real-time machine data, predict failures using machine learning, and displays key parameters through a user-friendly dashboard for efficient predictive maintenance



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4. Objectives:

- The primary objectives of the proposed framework are:
- a. To make the DAQ (Data Acquisition) device efficiently collect sensor data in real time.
 - b. To transmit the collected data in real time to a cloud platform running on a processing device.
 - c. To implement proper data processing and machine learning techniques for predicting potential failures.
 - d. To develop a custom dashboard that easily interprets data and sends notifications and alerts for timely action.

5. Methodology:

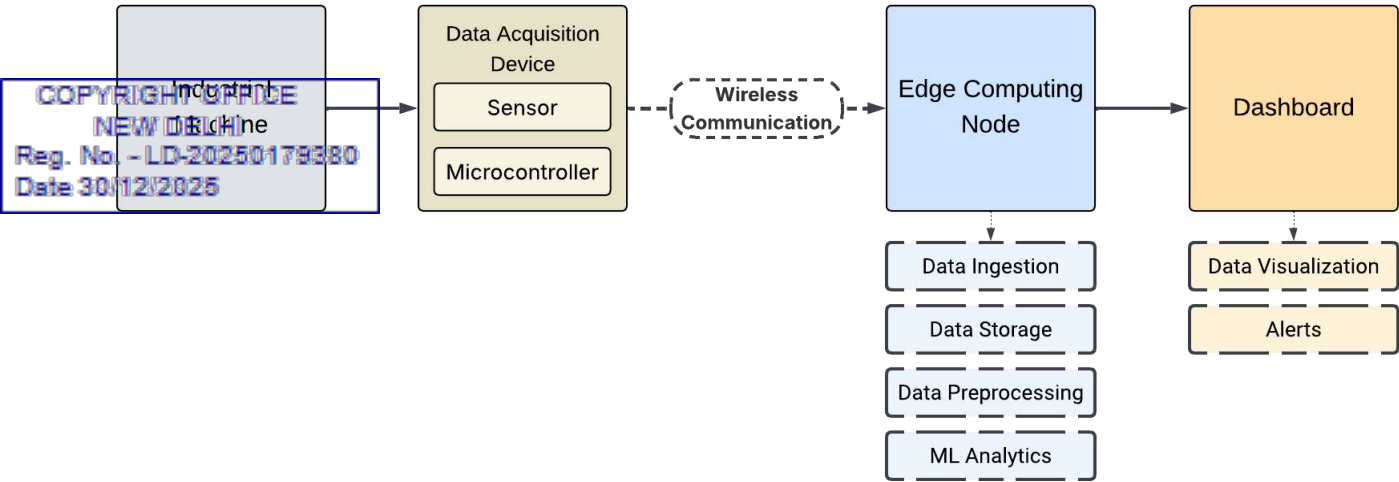


Figure 1: Block Diagram of Predictive Maintenance Framework

The proposed framework follows a systematic pipeline for real-time predictive maintenance, as illustrated in the block diagram. Each stage is designed to ensure accurate data acquisition, efficient transmission, intelligent processing, and user-friendly visualization.

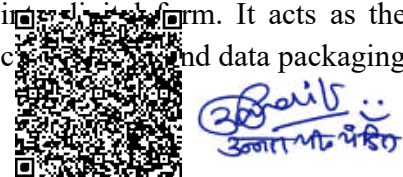
1. Industrial Machine:

The system begins with an industrial machine where various operational parameters (e.g., vibration, temperature, current, pressure) are continuously generated. These parameters serve as input signals for health monitoring.

2. Data Acquisition Device:

Sensors are deployed to measure machine parameters in real time. Depending on the application, accelerometers, temperature sensors, or current sensors may be used.

A Microcontroller (such as ESP32/Arduino/STM32) processes the raw sensor signals and converts them into a digital form. It acts as the primary Data Acquisition (DAQ) device, responsible for initial signal conditioning and data packaging.



3. Wireless Communication:

The processed data is transmitted wirelessly (using Wi-Fi, LoRa, or BLE depending on range and bandwidth requirements) from the DAQ device to the edge computing node. This ensures seamless, low-latency data transfer without manual intervention.

4. Edge Computing Node:

At the edge, data undergoes several stages:

- Data Ingestion : Receiving and collecting incoming streams of sensor data.
- Data Storage : Organizing and saving data locally or on a cloud-linked database for historical analysis.
- Data Preprocessing: Filtering noise, normalizing values, and preparing structured datasets suitable for ML analysis.
- ML Analytics: Applying machine learning algorithms to detect anomalies, classify machine states, and predict remaining useful life (RUL). This ensures predictive insights are generated in near real time.

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5. Dashboard:

The results of analysis are transmitted to a dashboard that provides two major functionalities:

- Data Visualization – Graphs, charts, and trend lines represent machine performance over time, enabling intuitive understanding of machine health.
- Alerts – Notifications are generated automatically when abnormal conditions are detected, supporting proactive maintenance decisions

This methodology ensures continuous, end-to-end monitoring, with seamless integration between sensors, microcontrollers, wireless communication, edge/cloud analytics, and visualization. The system operates in real time, making it scalable for diverse industrial applications

6. Conclusion:

This project showcases a predictive maintenance system based on IIoT that combines data gathering, wireless communication, and smart analysis for real-time monitoring of machine health. The system aids industries in transitioning from reactive approaches to proactive maintenance by delivering early fault detection and actionable insights via a tailored dashboard, thus reducing downtime and enhancing reliability.

The solution lowers maintenance expenses while simultaneously improving overall operational effectiveness. Featuring a flexible and scalable structure, the framework is suitable for various industrial sectors, facilitating Industry 4.0 efforts and data-oriented asset management.



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7. Advantages, Disadvantages, and Applications:

a. Advantages:

1. **Real-time monitoring:** The system enables continuous measurement of key machine parameters such as vibration, temperature, and current, ensuring that any abnormal deviation is captured instantly. This allows engineers to intervene at the earliest stage, reducing the risk of catastrophic failures.
2. **Reduced downtime:** By forecasting potential issues beforehand, companies can arrange maintenance during scheduled downtime instead of facing unexpected failures. This greatly minimizes idle machine hours, guaranteeing improved overall equipment effectiveness (OEE).
3. **Cost savings:** Predictive maintenance reduces unnecessary preventive checks and lowers the replacement of functioning parts. Eventually, this leads to improved use of spare parts, enhanced manpower efficiency, and decreased operating costs.
4. **Extended equipment lifespan:** Continuous monitoring helps maintain machinery within safe operational limits. Identifying minor issues before they escalate into severe damage extends the overall lifespan of costly industrial assets.
5. **Scalability and flexibility:** The framework can be implemented across multiple machines and industries, as it is based on modular DAQ systems and wireless data transfer. Additional sensors or algorithms can be integrated as per specific application requirements.
6. **Industry 4.0 alignment:** The system fits seamlessly into the vision of smart factories and digital transformation by combining IoT, machine learning, and cloud-edge platforms. It enables industries to adopt data-driven decision-making and enhance global competitiveness.

b. Disadvantages:

1. **Dependence on connectivity:** Since the framework relies on wireless transmission, weak signals, interference, or network outages can lead to delays in data transfer and analysis, reducing the reliability of real-time predictions.
2. **Data quality and volume requirements:** Machine learning models need large datasets that accurately represent machine behavior under different operating conditions. Poor-quality or insufficient data may reduce prediction accuracy and result in false alarms or missed failures.



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3. **Initial setup complexity:** Designing and integrating DAQ hardware, wireless modules, and dashboard software requires multidisciplinary expertise. For industries without skilled personnel, the initial deployment may be challenging.
4. **Sensor cost and maintenance:** High-precision sensors are necessary for reliable data collection. Deploying multiple sensors across large-scale industrial plants can be expensive, and maintaining sensor calibration adds additional costs.
5. **Cybersecurity risks:** Since data is transmitted wirelessly and stored on cloud platforms, the system may be vulnerable to cyberattacks, unauthorized access, or data breaches if strong security protocols are not implemented.



5. Applications:

1. **Smart factories and manufacturing units:** Predictive maintenance can be applied to monitor CNC machines, robotic arms, conveyor systems, and motors. Early fault detection ensures minimal production disruption and maximizes throughput.
2. **Power plants and energy systems:** Turbines, generators, transformers, and pumps are critical assets in energy plants. Continuous monitoring allows for early identification of overheating, vibration anomalies, and electrical faults, thereby preventing large-scale outages.
3. **Transportation and automotive industry:** Locomotives, aircraft engines, and heavy-duty vehicles can benefit from PdM by monitoring engine health, brake systems, and critical moving parts. This not only improves safety but also reduces operational delays.
4. **Oil and gas industry:** Equipment such as compressors, pumps, and drilling rigs operate under extreme conditions. Predictive maintenance ensures reliable operation by detecting early mechanical wear or leakage, thus reducing costly downtime and preventing hazardous incidents.
5. **Healthcare and medical equipment:** PdM frameworks can also be extended to sensitive hospital equipment such as MRI machines, ventilators, and robotic surgery devices, ensuring reliable functioning and patient safety.



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अनामिका सिंह

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ANNEXURE C
REVIEW PAPER

Nirikshak : An Industrial IoT and Predictive Maintenance Framework

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Abstract— Predictive maintenance (PdM) plays an important role in present-day industrial systems by enabling early identification of possible equipment failures, thereby reducing unplanned downtime, lowering maintenance expenses, and increasing asset lifespan. Recent advancements in the Industrial Internet of Things (IIoT) and machine learning (ML) have significantly strengthened this maintenance strategy by enabling continuous real-time data acquisition, intelligent pattern recognition, and data-driven maintenance planning across complex industrial infrastructures. This review summarizes recent progress in IIoT-based architectures, smart sensing technologies, and machine learning-driven analytics, while also highlighting emerging research directions that are influencing the evolution of predictive maintenance in smart manufacturing environments. Collectively, these technologies contribute toward achieving more dependable, efficient, and intelligent industrial operations. Furthermore, this paper introduces a conceptual IIoT-enabled predictive maintenance framework that combines real-time sensor data collection, edge-level processing, and machine learning analysis to facilitate effective condition monitoring within industrial systems.

Keyword— *Predictive Maintenance, Industrial Internet of Things (IIoT), Machine Learning, Smart Factories, Sensor Data Analytics, Edge Computing*

I. INTRODUCTION

The emergence of Industry 4.0 has transformed conventional manufacturing facilities into smart factories. A key driving force behind this transformation is the Industrial Internet of Things. It interconnects sensors, actuators, and industrial devices to enable seamless data acquisition, information sharing, and real-time equipment monitoring [1]. Through IIoT technologies, industries can automate operations, support data-driven decision-making, and enhance productivity via continuous monitoring and control

[2]. One of the major applications of IIoT in modern manufacturing is predictive maintenance. This approach enables early detection of potential equipment failures, thereby reducing downtime, lowering maintenance costs, and improving overall system efficiency [3]

Conventional maintenance strategies such as reactive maintenance, where repairs occur after failure, and preventive maintenance based on scheduled servicing, often result in unexpected outages, inefficient use of resources, and high maintenance costs[4]. In contrast, predictive maintenance relies on real-time sensor readings and operational logs from machine sensors and operation logs to find patterns that suggest possible faults or wear. This method is possible due to the integration of Machine Learning and data analytics, which allows for smart predictions and decisions in industrial settings[5]. Machine learning techniques including regression models, decision trees, support vector machines, and deep learning architectures can analyze large datasets for fault diagnosis and remaining useful life estimation[6]. These frameworks models, decision trees, support vector machines and deep learning networks, can analyze large data sets to spot anomalies, categorize fault types, and remaining useful life (RUL) of key machinery parts[3].

The effectiveness of predictive maintenance strongly depends on IIoT frameworks which provide the structure that is necessary for collecting usually include multiple layers like the perception layer consisting of sensors and microcontroller for data collection, network layer which use for data transmitting to processing unit, and application layer for data processing and visualization[1]. Preliminary data processing can occur locally at the edge, while more complex analytics and machine learning training happen in the cloud[2].

Despite its advantages, the integration of IIoT and ML presents several challenges. These include data quality issues, communication delays, cybersecurity threats and the difficulty of applying ML models across different industrial

settings[2]. Furthermore, choosing the right ML algorithms and IIoT architectures depends on specific application needs, such as the type of equipment, data collection rate and environmental factors. That is why understanding current IIoT frameworks and machine learning techniques is crucial for creating reliable and scalable predictive maintenance solutions[5].

This review aims to provide a comprehensive analysis of Industrial IoT frameworks and machine learning techniques which are used in predictive maintenance in smart factories and industrial machines. It discusses the basic concepts of IIoT architecture, explores different data collection and communication methods and looks at the role of machine learning algorithms in detecting and predicting faults. It also points out the advantages, limitations, and new research trends in this field, providing valuable insights for researchers and industry professionals to design effective, smart and durable predictive maintenance systems for the future of smart manufacturing[2], [6].

II. INDUSTRIAL IOT (IIOT) FRAMEWORKS FOR SMART FACTORIES

A. Overview of IOT in Industry

The Industrial Internet of Things (IIoT) integrates information technology with operational technology by connecting sensors, actuators, and intelligent devices into a unified industrial network. These interconnected components continuously acquire, process, and analyze real-time data, enabling improved automation, enhanced decision-making, and more efficient manufacturing operations. [7], [8].

IIoT enhances conventional manufacturing systems by integrating smart sensing devices with cloud and edge computing technologies. This architecture supports continuous real-time monitoring, enables predictive data analysis, and facilitates automated decision-making processes within industrial environments. [7], [9].

Basically, IIoT forms the backbone of Industry 4.0, supporting intelligent manufacturing. It connects machines, people, and digital systems to create cyber-physical production environments systems that can sense, respond and optimize themselves automatically[7], [10].

Key Benefits of IIoT:

- **Cost Reduction:** IIoT helps lower maintenance and operational costs by using predictive *maintenance*. By continuously analyzing sensor data, systems can detect faults early before they cause costly breakdowns. This process not only prevents failures but also reduces unnecessary data transmission and computing load, saving resources in the long run[7], [10].
- **Minimised Downtime:** With continuous condition monitoring, IIoT systems can identify early signs of mechanical issues such as vibration imbalance or wear. Detecting these problems early helps prevent unplanned Machine failures which increasing

machine availability and improving Overall Equipment Effectiveness[8], [9].

- **Real-Time Monitoring:** Smart sensors continuously monitor critical performance parameters such as temperature, vibration, and pressure in real time. Edge devices process this data locally, sending only critical information to the cloud. This creates a fast feedback loop between machines and analytics systems which allowing instant response and fine tuning of operations[7], [11].

B. IIoT Architectures

A basic IIoT framework usually works in three layers devices that sense data, networks that send it and cloud systems that analyze it, This architecture simplifies the efficient collection, transmission, and intelligent utilization of data.

- **Perception Layer (Device Layer):** This layer serves as the initial stage of the system and consists of sensors and microcontrollers responsible for acquiring physical parameters such as temperature, vibration, and electrical current from the real-world environment. The sensors detect changes around them and send this information to small boards like Arduino or ESP32 for initial processing. These devices convert the data into digital form so that it can be shared easily[7], [8].
- **Communication Layer:** This layer acts like a bridge that connects devices to the cloud. It uses communication technologies such as Wi-Fi, Ethernet, or cellular networks (4G/5G) to transfer data. Protocols like MQTT or HTTP are used to make sure the data reaches safely and quickly. Sometimes, a gateway is used to filter or clean the data before storing it to the cloud[9], [11].
- **Cloud Layer:** This is where the real intelligence happens. The data coming from devices is stored and analyzed using cloud platforms like AWS or Azure AI models or algorithms study the data to find trends, detect problems or predict failures. The results are shown on dashboards or mobile apps so that users can easily monitor and make decisions[8], [10].

III. PREDICTIVE MAINTENANCE TECHNIQUES:

A. Maintenance Strategies Evolution:

Maintenance practices in industrial environments have evolved significantly over the years to meet the increasing demands for efficiency, reliability, and cost control. The earliest approach, known as reactive maintenance (or corrective maintenance), involved fixing or replacing equipment only after a breakdown occurred. Although this method required little upfront planning or investment, it often resulted in unexpected production stoppages, expensive repairs and safety hazards.

To mitigate these issues, preventive maintenance strategies were introduced. This strategy plans maintenance actions at

predefined intervals based on operating hours, usage patterns, or manufacturer guidelines, without considering the actual condition of the equipment. While preventive maintenance helps reduce the risk of sudden breakdowns, it often results in over-maintenance, where parts are serviced or replaced unnecessarily. This not only increases costs but can also lead to wasting resources.

As industrial systems increased in complexity, the need for more intelligent and adaptive maintenance strategies became evident. This led to the introduction of condition-based maintenance (CBM) and later predictive maintenance approaches. CBM utilizes continuous monitoring of machine health indicators such as vibration, temperature, and pressure to determine appropriate maintenance timing. Predictive maintenance extends this concept by employing data analytics and machine learning techniques to anticipate potential failures before they occur. [12]

B. Types of Maintenance:

Industrial maintenance strategies are essential for ensuring equipment reliability, operational efficiency, and cost control. The four maintenance strategies include Reactive (Run-to-Failure), Preventive (Time-Based), Condition-Based Maintenance (CBM), and Predictive Maintenance (PDM).

Reactive maintenance, commonly referred to as the run-to-failure approach, the most basic form of maintenance. It involves performing repairs only after equipment failure occurs. Although this strategy reduces initial maintenance expenditure, it frequently leads to unplanned downtime and increased long-term costs caused by sudden failures and equipment damage [3].

Preventive maintenance follows a structured schedule in which servicing is performed at predefined time or usage intervals according to manufacturer recommendations, irrespective of the actual equipment condition. This approach helps minimize unexpected breakdowns and contributes to extending the operational lifespan of assets. [13].

Condition-Based Maintenance (CBM) offers a more flexible alternative. It relies on real-time or periodic monitoring of equipment health using real-time sensor data such as vibration, temperature, pressure Maintenance is carried out only when the data indicates that performance is declining[14].

Predictive Maintenance (PdM) is the most advanced form of data-driven maintenance strategy. It employs machine learning and data analytics to anticipate potential equipment failures before they occur. By evaluating both historical patterns and real-time operational data, PdM allows maintenance teams to schedule interventions precisely when required. This approach minimizes unplanned downtime, lowers maintenance costs, extends asset life, and enhances overall system reliability. [12].

C. Data-Driven Predictive Maintenance

Data-driven PdM lies at the intersection of sensor technology, IoT, and intelligent decision-making. Its workflow typically includes three key stages: data collection,

preprocessing, and analysis, which together convert raw sensor data into data-driven actions[3].

Data collection consists of sensors that continuously capture data such as vibration, temperature, pressure, current, and voltage. Through IoT networks and edge computing gateways, this information is transmitted in real time for monitoring and storage[14].

Pre-processing means processing before analysis, raw data undergoes cleaning, normalization, and feature extraction using methods like statistical filtering, Fourier transforms, and wavelet analysis. This step removes noise and enhances the clarity of degradation patterns[13].

In analysis and modelling Machine learning algorithms are applied to identify correlations, anomalies, and failure. Integration with cloud platforms allows scalable data storage and real-time data analytics, enabling remote monitoring and rapid decision support[12].

By moving away from fixed maintenance, data-driven PdM supports digital twins, remote supervision, and automated decision-making effectively shifting maintenance from a reactive response to a proactive strategy.

D. Machine Learning Models for Predictive Maintenance

Machine learning (ML) plays a vital role in predictive maintenance (PdM) systems. By learning from historical and real-time data, ML models improve fault diagnosis, detect early signs of degradation, and enable accurate prediction of equipment failures. Broadly, these models can be categorized into supervised and unsupervised learning approaches[3].

Supervised learning is applied when labelled datasets are available meaning the system has examples of both normal and faulty conditions to learn from

Regression models are used to estimate variables such as remaining useful life (RUL) or wear levels. While they are simple to implement and interpret, their accuracy can decline when dealing with nonlinear degradation patterns.

Random Forest (RF) models are effective for handling noisy, multidimensional data and are relatively robust against overfitting. However, they can be intensive, especially when processing large datasets[12].

Support Vector Machines (SVMs) perform well with small to medium-sized datasets and are particularly suited for fault classification. Their effectiveness depends heavily on careful selection and tuning of kernel parameters.

Artificial Neural Networks (ANNs), including deep neural networks (DNNs), are powerful tools for capturing complex nonlinear relationships in sensor-rich environments. Despite their high predictive accuracy, they often lack interpretability[3].

Unsupervised learning techniques are used when labelled data are unavailable. These models aim to identify hidden patterns or anomalies directly from unlabelled data.

Clustering Algorithms: Techniques such as *k-means* and *Gaussian Mixture Models* group similar operational behaviors, helping to uncover latent machine states and detect emerging fault conditions.

Anomaly Detection Methods: These approaches identify deviations from normal operating patterns using probabilistic thresholds or reconstruction errors, enabling early fault recognition.

Deep Learning Architectures: Models such as Autoencoders, Convolutional Neural Networks (CNNs), and Long Short-Term Memory networks (LSTMs) are particularly effective for analyzing high-dimensional, time-series data like vibration signals. They can learn temporal dependencies and degradation sequences that precede equipment failure.

Hybrid and Ensemble Approaches: Recent advancements have led to hybrid models that combine the strengths of multiple learning algorithms. For example, integrating CNNs with Bayesian inference can enhance both diagnostic accuracy and interpretability. Such ensemble systems are increasingly being deployed in critical sectors including energy, transportation, and chemical processing, where reliability and precision are paramount.

IV. INTEGRATION OF IIOT AND MACHINE LEARNING FOR PREDICTIVE MAINTENANCE

The combination of the Industrial Internet of Things (IIoT) and machine learning (ML) is changing how industries approach predictive maintenance (PdM). Instead of reacting to failures or following fixed maintenance schedules, companies can now use data-driven methods to predict when equipment might fail and act in advance.

IIoT networks make this possible by connecting large numbers of sensors that continuously collect data such as vibration, temperature, and pressure from industrial machines. These real-time data streams are then analyzed by ML and deep learning (DL) models to identify patterns of wear and potential faults. As a result, maintenance can be scheduled only when needed, which improves equipment reliability, reduces downtime, and saves costs in industries like manufacturing, energy, and transportation[14].

A. Sensor Data Transmission and Processing

Sensor data are the foundation of IIoT-based predictive maintenance systems. Sensors placed on equipment such as motors, pumps, or conveyor belts collect continuous data about operating conditions. These readings are sent to data processing systems through communication protocols like MQTT, OPC-UA, or HTTP/S, which ensure fast and reliable data transfer between devices, gateways, and cloud platforms[10]. A typical data flow in PdM systems includes the following steps:

- **Data collection:** Smart sensors record signals such as vibration, sound, and temperature.

- **Pre-processing at the edge:** The data are cleaned and simplified (for example, by removing noise or normalizing readings) to make analysis faster.
- **Data transmission to the cloud:** Important information is sent to cloud platforms for long-term storage and advanced analysis.
- **Real-time synchronization:** Tools like Apache Kafka and Spark help share and update data between factory-floor systems and central databases in real time[15].

B. Edge vs. Cloud Processing of PdM Models

Modern PdM systems often use a hybrid approach, combining both edge and cloud computing to make the most of each technology

- **Edge Processing:** In edge computing, small processors or gateway devices located close to the machines handle quick, on-site tasks such as identifying sudden vibrations or temperature spikes. Because the data are processed locally, response times are very fast, which is crucial for preventing damage for example, automatically stopping a motor when a serious fault is detected[15].
- **Cloud Processing:** Cloud computing, on the other hand, is used for more complex tasks that require higher computing power. The cloud stores large amounts of historical data, trains deep learning models, and compares data from multiple sites or machines. It also manages updates to models, which can then be sent back to edge devices for real-time use[14].

By combining both edge and cloud processing, industries can ensure fast responses to faults while still benefiting from large-scale data analytics and long-term learning[10].

C. Real-Time Monitoring Dashboards and Alert Systems

Real-time monitoring and alert systems are key features of IIoT-based PdM solutions. These systems turn complex sensor data into clear, easy-to-understand visuals for maintenance teams. Dashboards show information like current equipment status, performance trends, and predicted failure times, helping teams make quick and informed decisions[12].

Users can set their own alert thresholds so that when a sensor detects something abnormal, the system automatically sends notifications via email, SMS, or even directly to other machines. Many systems are also connected to Computerized Maintenance Management Systems (CMMS), which automatically create maintenance tickets or work orders when needed.

Common tools used for visualization include Grafana, Power BI, and various custom web platforms that connect directly to both edge and cloud-based ML systems[15].

V. PROPOSED SYSTEM

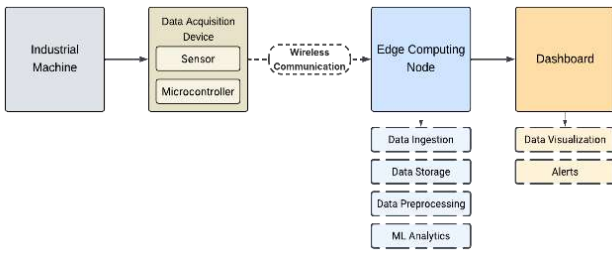


Figure 1: Block Diagram of proposed Predictive Maintenance Framework

The proposed system for predictive maintenance in industrial environments integrates Industrial Internet of Things frameworks with machine learning analytics to monitor, process, and analyze real-time data from industrial machines. The block diagram in Figure 1, consists of four major components: Industrial Machine, Data Acquisition Device, Edge Computing Node and Dashboard Interface. This section elaborates on the functional flow, components involved and data processing stages which helps for predictive maintenance.

A. Industrial Machine

The process starts with the industrial machine which serves as the primary data source. A machine can be a motor, pump, conveyor or compressor which is equipped with various sensors to capture operational parameters like temperature, vibration, pressure, current or rotational speed. The condition of the machine is continuously monitored to identify any abnormal patterns or deviations that might indicate wear, imbalance, or component degradation.

Each machine functions as a cyber-physical system (CPS) that integrates physical processes with computational control. The sensor data collected represents the real-time condition of the equipment and serves as the basis for predictive analytics. Connecting these machines to a data acquisition layer enables transparency, traceability, and intelligent monitoring within the industrial environment.

B. Data Acquisition Device

The data acquisition (DAQ) device serves as mediator between the machine and the computational infrastructure. It typically consists of sensors and a microcontroller or embedded processor.

a. Sensors

A sensor is an electronic device that detects physical quantities such as temperature, pressure, vibration etc. and converts them into electrical signals that can be measured and analyzed. Different types of sensors are used depending on the nature of the machine and the type of fault to be detected. Following are examples of various sensors-

- Vibration (accelerometer) sensors are used to detect mechanical faults such as imbalance, misalignment, and bearing defects.
- Temperature sensors monitor overheating caused by friction, overload, or insulation failure.

- Current and voltage sensors track electrical abnormalities in motors and power circuits.
- Pressure and flow sensors supervise variations in fluid-based system operations.

These sensors generate analog or digital signals corresponding to the machine's physical parameters.

b. Microcontroller Unit

A microcontroller such as an Arduino, ESP32, STM32 or Raspberry Pi Pico acts as the local processing and control unit. It is responsible for sampling sensor data at predefined intervals, performing analog-to-digital conversion and formatting the data for transmission.

The microcontroller may also perform initial filtering operations such as noise removal using low-pass filters to ensure clean and accurate sensor readings.

In the proposed framework, the microcontroller transferred the processed data wirelessly to the Edge Computing Node using communication protocols such as Wi-Fi, Bluetooth, LoRa or Zigbee depending on the application environment and range requirements.

C. Edge Computing Node

The Edge Computing Node serves as an important component of the system which enables local data processing, analytics, and machine-learning-based decision-making. Instead of transmitting all sensor data to a centralized cloud server, edge computing allows preliminary analysis to occur closer to the data source. This architecture minimizes latency, reduces bandwidth usage, enhances system reliability, and ensures faster real-time responses for predictive maintenance applications. The edge node performs the following key functions-

a. Data Ingestion

Raw sensor data from the microcontroller is collected and stored properly for further processing. Data ingestion modules manage the gathering of real-time streaming data and store it in temporary buffers or databases.

b. Data Storage

The edge device keeps a local database that temporarily holds time-series data. This enables the use of historical records for short-term analysis and training of local models.

c. Data Pre-processing

Before applying machine learning algorithms, the collected data goes through several pre-processing steps to improve its quality and readiness for analysis. First noise filtering removes unwanted or irrelevant components from sensor signals. This ensures cleaner input for analysis. Then the data is normalized or scaled to standardize measurement ranges and help models converge more effectively. After that feature extraction techniques are used to derive meaningful parameters such as mean, root mean square (RMS), kurtosis and frequency domain features using methods like the Fast Fourier Transform (FFT). These extracted features capture important statistical and spectral characteristics. This helps assess machine health conditions more accurately and boosts the predictive ability of the learning models.

d. Machine Learning Analytics

The pre-processed data is then used by machine learning algorithms operating at the edge node for smart analysis and decision-making. Depending on the application, various types of algorithms can be applied. Classification algorithms such as Decision Trees, Support Vector Machines (SVM) and Random Forests are commonly used for fault detection and diagnosis. They categorize machine states based on extracted features. Regression-based algorithms like Linear Regression and Gradient Boosting estimate continuous variables such as remaining useful life (RUL) or the degradation levels of components. The trained models provide near real-time predictions of potential faults, and the results are sent to the monitoring dashboard for visualization and alert generation.

D. Wireless Communication

Data transfer between the microcontroller and the edge node happens wirelessly. This approach ensures flexibility and scalability in industrial settings. Common wireless communication technologies include Wi-Fi (IEEE 802.11), which supports short- to medium-range, high-data-rate transmission. This makes it suitable for real-time monitoring applications. LoRa enables long-range communication with low power consumption. It is ideal for remote or battery-powered systems. Bluetooth Low Energy (BLE) offers cost-effective, short-range data transfer for compact setups.

The communication layer uses lightweight and efficient protocols like MQTT (Message Queuing Telemetry Transport) and HTTP REST APIs. These protocols provide reliable data exchange between devices. They also offer high efficiency, fault tolerance, and smooth integration within Industrial IoT frameworks.

E. Dashboard and Visualization

The Dashboard Interface serves as the human-machine interaction layer of the system. It provides an easy-to-understand visualization of real-time machine status, performance metrics, and predictive insights generated by the ML analytics engine.

a. Data Visualization

The dashboard displays graphical representations of key parameters such as temperature, vibration and health scores. These visualizations help maintenance operator to quickly interpret machine conditions and operational patterns.

b. Alerts and Notifications

Whenever the edge analytics module detects abnormal behavior or predicts an upcoming failure, alert messages or notification is sent to dashboard as well as to the mobile phone of the maintenance operator.

By incorporating visual dashboards, the system ensures transparency and assists operators in making timely, data-driven maintenance decisions.

VI. CHALLENGES

A. Data Quality and Sensor Reliability:

In the world of industrial IoT (IIoT), reliable and accurate sensor data is the foundation for effective predictive maintenance. Machine learning models depend on clean, high-quality data to detect unusual behavior and predict equipment failures. But keeping sensor data consistent and trustworthy isn't always easy. Wireless sensors can face interference from other signals, changing environmental conditions, or even physical damage and cyber tampering. Over time, sensors can drift out of calibration, go offline, or record incorrect readings, which can lead to missing or misleading data. To maintain data integrity, companies need to regularly check and calibrate their sensors, use strong communication protocols, and include backup or redundant sensors where possible. Data cleaning and pre-processing also play an important role in removing noise and errors before the data is analyzed[14].

B. Cybersecurity Concerns in IIoT:

As factories and industrial systems become more connected, cybersecurity is becoming a top priority. IIoT systems handle sensitive data and control critical equipment, which makes them attractive targets for hackers. Attacks can take many forms—data theft, ransomware, or even deliberate interference with operations. The problem is made worse because many IIoT devices, such as small sensors or actuators, have limited computing power and can't support advanced security features. To protect these systems, security needs to be built in at every level. This includes encrypting data, using secure boot mechanisms, keeping software up to date, and adding hardware-based security features where possible. Artificial intelligence can also help detect attacks in real time. In addition, privacy-preserving approaches like federated learning and "zero trust" security frameworks ensure that every user and device must be verified before gaining access to the system[10].

C. Scalability of Machine Learning Models:

One of the biggest challenges in IIoT is scaling machine learning models to handle the massive amounts of data coming from thousands of sensors in large industrial plants. Cloud and edge computing have made it easier to process this data, but the high speed and variety of incoming information can still overwhelm existing systems. To manage this, companies are turning to distributed or federated learning, which allows models to learn from data stored in different locations without needing to move it all to one place. Models also need to be updated automatically as new data arrives to stay accurate. Managing real-time analysis across both cloud and edge devices is especially important for time-sensitive operations like predicting equipment failures.

Technologies such as Kubernetes and hybrid computing setups help by balancing workloads across multiple systems. However, research shows that even advanced solutions can struggle when scaled to hundreds of thousands of connected devices, showing that scalability remains an on-going challenge[11].

VII. FUTURE TRENDS

As Industrial Internet of Things (IIoT) technologies and predictive maintenance systems continue to develop, several transformative trends are emerging that will shape the future of smart manufacturing and intelligent asset management. These advancements promise to improve operational efficiency, reduce downtime, and enable autonomous decision-making in industrial settings.

A. AI-Driven Self-Healing Factories:

The idea of self-healing factories represents a major shift from reactive maintenance to fully autonomous systems that can detect, diagnose, and recover from failures without human help. AI-powered self-healing mechanisms combine predictive maintenance with automated recovery loops. This allows for real-time fault detection, dynamic load balancing, and autonomous corrective actions. When a component shows signs of wear, AI systems can redistribute tasks to working equipment, trigger on-demand production of replacement parts using additive manufacturing, or adjust operating parameters to compensate for partial failures while minimizing downtime. Reinforcement learning models continuously improve response strategies based on historical machine behavior, enabling factories to evolve and adjust on their own[16], [17].

B. Digital Twins for Predictive Maintenance:

Digital twin technology has become a crucial tool for advanced predictive maintenance strategies. A digital twin creates a dynamic virtual replica of physical assets that is regularly updated with real-time sensor data, allowing manufacturers to monitor equipment health, simulate maintenance scenarios, and predict failures before they happen. By combining IoT sensor networks with AI-driven analytics, digital twins allow for accurate forecasting of equipment wear, optimal maintenance scheduling, and scenario testing without disrupting actual production. This proactive approach shifts maintenance from time-based schedules to data-driven, condition-based strategies, greatly improving asset reliability and lowering operational costs[18], [19].

C. 5G and Industrial IoT Connectivity Improvements:

The rollout of 5G technology is transforming IIoT connectivity by offering ultra-high bandwidth, ultra-low latency, and large device connectivity capabilities that are crucial for real-time industrial applications. Private 5G networks provide manufacturers with a dedicated, secure communication system with performance guarantees that are essential for time-sensitive operations like predictive maintenance, remote equipment control, and autonomous systems. With the ability to support millions of connected devices per square kilometre, 5G removes the need for complex cabling while delivering the reliability and low-latency communication necessary for critical manufacturing processes. The combination of 5G with edge computing further speeds up real-time data processing and AI-driven decision-making at the network edge[20].

D. Hybrid Edge-Cloud Architectures:

Modern IIoT systems are increasingly using hybrid edge-cloud architectures that combine the strengths of both models to optimize performance, cost, and functionality. In these setups, edge computing handles time-critical tasks like real-time anomaly detection, equipment monitoring, and immediate control responses, while cloud platforms manage large-scale data collection, long-term analytics, enterprise-wide coordination, and heavy model training. This complementary method ensures quick responses on the factory floor while also providing strategic oversight and thorough data analysis capabilities at the organizational level. New techniques like federated learning enable collaborative model training across distributed edge devices without centralizing sensitive data, maintaining privacy while improving predictive accuracy[20], [21].

VIII. CONCLUSION

This review brings together the latest progress in predictive maintenance (PdM) made possible by combining the Industrial Internet of Things (IIoT) with machine learning (ML). The findings show that this combination allows industries to detect problems in real time, improve equipment reliability, and lower maintenance costs. By using smart sensors, data analytics, and connected digital systems, companies can predict faults earlier, plan maintenance better, and avoid unexpected downtime[12].

The main advantage of linking IIoT and ML is that it helps industries move from traditional, routine maintenance toward a more proactive and flexible approach. This makes maintenance smarter, faster, and more accurate, supporting better decision-making and improving efficiency as industries move toward the goals of Industry 4.0[3].

For the future, research should focus on improving data quality and sensor performance, developing ML models that are easier to understand and use, and creating shared standards so that different systems and devices can work together securely. Cybersecurity and data privacy also need to be top priorities to keep these systems safe[14].

Finally, it's important to take a human-centered approach that combines new technology with real-world experience. Working together across fields will help turn research into practical tools that industries can use every day, leading to smarter, safer, and more reliable factories in the future[12].

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ANNEXURE D
RESEARCH PAPER

Nirikshak: An Industrial IoT & Predictive Maintenance Framework

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Keywords: Industrial IoT, Predictive Maintenance, Edge Computing, Machine Learning, MQTT Communication, Anomaly Detection, Fault Prediction, Remaining Useful Life (RUL), Smart Manufacturing, Real-Time Monitoring

Abstract: Predictive maintenance (PdM) has emerged as a critical approach for reducing unplanned downtime and improving operational efficiency in industrial systems. However, the adoption of PdM remains limited, particularly among small and medium-sized enterprises (SMEs), due to the high cost, proprietary nature, and implementation complexity of existing commercial solutions. These systems often require significant capital investment and specialized expertise, creating a substantial barrier to entry. This paper presents a low-cost, scalable Industrial IoT-based predictive maintenance system that integrates multi-modal sensors, MQTT-based communication, and machine learning techniques for real-time condition monitoring and fault prediction. The proposed system utilizes an embedded edge device for data acquisition and preprocessing, while cloud-based analytics enable anomaly detection and fault classification. Experimental evaluation demonstrates a fault detection accuracy of 94.2%, with an end-to-end system latency of 320 ms. The total system cost is approximately Rs.8,500, representing a significant reduction compared to conventional PdM solutions. The proposed approach offers an affordable and practical alternative for SMEs, enabling early fault detection, reduced maintenance costs, and improved equipment reliability. This work contributes toward democratizing predictive maintenance through accessible, open, and efficient technological solutions.

1 INTRODUCTION

Modern industrial operations rely heavily on continuous and reliable machine operation to maintain productivity and reduce operational costs. Predictive maintenance (PdM) has emerged as an effective strategy to reduce equipment failures and optimize maintenance schedules (Theissler et al., 2021), (Samatas et al., 2021). However, despite its advantages, PdM adoption remains limited, particularly among small and medium-sized enterprises (SMEs), due to the high cost, proprietary nature, and technical complexity of existing commercial solutions.

Traditional maintenance strategies are broadly categorized into reactive and preventive approaches. Reactive maintenance, also known as run-to-failure, allows equipment to operate until breakdown, leading to unplanned downtime, high repair costs, and potential safety risks. Preventive maintenance, on the other hand, relies on fixed maintenance schedules regard-

less of the actual condition of the equipment, often resulting in unnecessary servicing, increased operational expenses, and inefficient utilization of component life.

Although predictive maintenance addresses these limitations by enabling condition-based monitoring and early fault detection (Aissani et al., 2024), (Benhanifia et al., 2025), current implementations are often expensive and lack accessibility. Moreover, many existing research efforts focus primarily on isolated machine learning models without providing a complete, real-time, and deployable system architecture.

To address these challenges, this paper proposes a low-cost, end-to-end IoT-based predictive maintenance system that integrates embedded edge devices (STM32), multi-modal sensors, MQTT-based communication, and machine learning algorithms (Elkateb et al., 2024), (Hamasha et al., 2025). The system is designed to provide real-time monitoring, anomaly detection, and fault prediction while maintaining afford-

ability and ease of deployment, making it suitable for SMEs and practical industrial applications.

2 LITERATURE REVIEW

2.1 Industrial IoT Architectures

Industrial IoT (IIoT)-based predictive maintenance systems are typically built using three architectural paradigms: cloud-based, edge-based, and hybrid architectures(Andriulo et al., 2024),(Savaglio et al., 2024). Cloud-centric systems rely on centralized processing, offering high computational power for complex analytics but suffering from high latency and bandwidth dependency. In contrast, edge-based architectures perform data processing near the data source, enabling low-latency responses and reduced network load. However, they are constrained by limited computational resources.

To overcome these limitations, hybrid edge–cloud architectures have emerged as the preferred approach(Andriulo et al., 2024),(Hamasha et al., 2025), combining real-time processing at the edge with scalable analytics in the cloud. This distributed model improves responsiveness while maintaining computational efficiency and scalability . Despite these advancements, implementing such architectures in a cost-effective and easily deployable manner remains a challenge.

2.2 Machine Learning Models for Predictive Maintenance

Machine learning (ML) plays a central role in predictive maintenance by analyzing sensor data to detect anomalies and predict failures(Theissler et al., 2021),(Taşcı et al., 2023),(Elkateb et al., 2024). Common supervised learning models include Random Forest, Support Vector Machines (SVM), and deep learning models such as Long Short-Term Memory (LSTM) networks(Taşcı et al., 2023),(Rakholia et al., 2025).

Random Forest models are widely used for fault classification due to their robustness and ability to handle nonlinear data. SVMs are effective for binary classification problems, especially with limited datasets. LSTM networks, on the other hand, are particularly suitable for time-series data(Taşcı et al., 2023), capturing temporal dependencies in sensor signals and achieving high prediction accuracy. However, deep learning models often require large datasets and high computational resources, making them less suitable for real-time embedded applications.

2.3 Commercial Systems vs Research Prototypes

Commercial predictive maintenance solutions, such as those offered by major industrial vendors, provide complete end-to-end platforms with integrated hardware, analytics, and visualization tools(Aissani et al., 2024),(Benhanifia et al., 2025). These systems typically achieve high accuracy and reliability but are associated with significant costs, often making them inaccessible to SMEs. Additionally, they are usually proprietary, limiting customization and scalability.

In contrast, academic and research-based systems primarily focus on developing and evaluating ML models or specific components of PdM systems(Theissler et al., 2021),(Benhanifia et al., 2025). While these approaches demonstrate promising accuracy and innovative techniques, they often lack full system integration, real-world validation, and deployability. Many research works emphasize algorithmic performance without addressing communication, real-time constraints, or user interface design.

2.4 Limitations of Existing Work

Despite significant progress, existing predictive maintenance solutions exhibit several limitations:

- **High Cost:** Commercial systems require substantial investment, restricting adoption in SMEs(Aissani et al., 2024),(Benhanifia et al., 2025).
- **Lack of Integration:** Research prototypes often focus on isolated components rather than complete end-to-end systems.
- **Limited Real-Time Capability:** Cloud-dependent architectures introduce latency, limiting real-time fault detection.
- **Complex Implementation:** Integration of IoT, ML, and communication systems requires significant expertise and infrastructure(Samatas et al., 2021),(Hamasha et al., 2025).

2.5 Research Gap

From the literature, it is evident that there is a lack of affordable, fully integrated, and real-time predictive maintenance systems suitable for practical deployment in SMEs(Benhanifia et al., 2025),(Hamasha et al., 2025). Existing solutions either provide high performance at high cost (commercial systems) or low-cost but incomplete implementations (research prototypes).

Therefore, there is a clear need for a cost-effective, end-to-end IoT-based predictive maintenance system that integrates sensing, edge processing, communication, and machine learning while ensuring real-time performance and ease of deployment.

3 Methodology

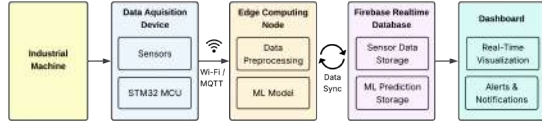


Figure 1: System Architecture

The suggested Nirikshak scheme involves the use of an Industrial IoT-driven predictive maintenance platform that includes sensor-based data collection, edge computing, MQTT protocol for data communication, machine learning algorithms for data analytics, cloud computing for storing data, and visualization of results via dashboards. The platform enables continuous monitoring of industrial machines and real-time fault detection based on a well-designed end-to-end solution pipeline.

3.1 Industrial Machine

The architecture begins with industrial machines like AC motors, compressors, pumps, and rotary machines. During their functioning, the machines create some physical and electrical variables such as vibration, temperature, rotation, change in voltage, and change in current among others. The variables serve as the indices of the state of the machines.

Monitoring the behavior of the machines continuously enables the recognition of unusual machine operation, which is related to factors such as bearing wearing, overheating, misalignment of the shaft, overload, and wear among others. Since most industrial problems occur gradually, it becomes easier to detect the faults at an early stage.

3.2 Data Acquisition Device

The process of collecting data uses a few sensors connected to an STM32 embedded controller system. A multiplicity of sensors is used in order to address the fact that faulty situations have an effect on different variables.

The analog sensor signals are processed using amplifiers and low-pass filters before being digitized us-

Table 1: Sensor Components Used

Sensor Type	Model	Description
Temperature	DS18B20	Temperature sensing
Vibration	MPU6050	Vibration monitoring
RPM	Proximity Sensor	Speed measurement
Voltage	ZMPT101B	Voltage monitoring
Current	SCT-013-030	Current monitoring

ing an ADC. Signal processing reduces external interference and improves the accuracy of the ADC. The microcontroller continuously captures the sensor signal values at specified times and holds the captured values in buffers.

Table 2: Hardware Specifications of Data Acquisition Unit

Component	Specification	Purpose
MCU	STM32F103	DAQ
Core	Cortex-M3	Processing
Clock	72 MHz	Sampling
ADC	12-bit	Conversion
RAM	20 KB	Buffering
Flash	64 KB	Storage
WiFi	ESP32	Communication
Supply	5V DC	Operation

Preprocessing such as noise filtering, normalization, and sensor calibration is done by the STM32 microcontroller before transferring the data. Preprocessing helps reduce communication overhead.

3.3 MQTT-Based Communication

The processed sensor data is transmitted wirelessly using the MQTT communication protocol. MQTT follows a lightweight publish-subscribe architecture and is widely adopted in Industrial IoT systems due to its low bandwidth utilization, low power consumption, and efficient real-time communication capability.

In the proposed implementation, the STM32-ESP32-based embedded device acts as the MQTT publisher and continuously publishes sensor readings in JSON format to the MQTT broker. The Mosquitto MQTT broker manages message distribution between connected devices. A Python-based edge computing application acts as the MQTT subscriber and continuously receives live sensor data for preprocessing, machine learning inference, and cloud synchronization.

Figure 2 illustrates the MQTT-based communication architecture used in the proposed system.

The MQTT-based communication layer enables efficient real-time transmission between the acquisition device and processing layer with minimal latency

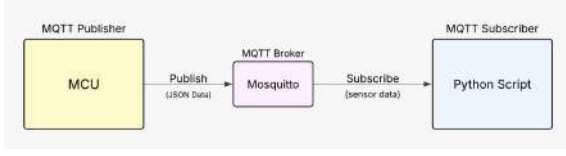


Figure 2: System Architecture

Table 3: Communication Configuration Parameters

Parameter	Value
Protocol	MQTT
QoS Level	1
Publish Interval	500 ms
Communication Medium	WiFi
Data Format	JSON
Topic Structure	factory/machine01/sensors

and network overhead.

3.4 Edge Computing Node

The edge computing node is implemented locally on a laptop system using Python and Visual Studio Code. The edge node acts as the primary intelligence layer of the proposed predictive maintenance framework (Artiushenko et al., 2024), (Savaglio et al., 2024). It performs data preprocessing, dataset generation, machine learning inference, anomaly detection, Remaining Useful Life (RUL) prediction, cloud synchronization, and dashboard integration.

The edge computing architecture consists of four major modules including the MQTT subscriber module, preprocessing module, machine learning inference module, and Firebase cloud integration module.

3.4.1 MQTT Subscriber Module

The MQTT subscriber module continuously receives live sensor data transmitted from the ESP32-based DAQ system. The subscriber application parses incoming JSON packets and performs real-time logging of sensor readings into CSV files. These CSV datasets are later utilized for machine learning model training, historical analysis, and predictive maintenance research.

3.4.2 Dataset Collection and Preparation

Real-time sensor data captured from the IIoT setup with and without faults is used as input to the proposed model. Around 10000 samples have been obtained from sensors such as temperature, vibration, RPM, voltage, and current with the help of a data acquisition system based on STM32 technology. The

preprocessing of the data includes steps like normalization, noise reduction, and feature extraction using the sliding window approach before being fed to the machine learning algorithm. An 80-20 split was performed for training and testing with 5-fold cross-validation.

3.4.3 Data Preprocessing Module

The preprocessing module processes incoming raw sensor readings before machine learning analysis. The preprocessing pipeline includes noise filtering, missing value handling, normalization, signal conditioning, feature extraction, and data formatting.

Sliding window techniques are used to convert continuous sensor signals into feature vectors suitable for machine learning inference. Statistical and time-domain features are extracted from the sensor data to identify degradation patterns and abnormal machine behavior.

The extracted statistical features include mean, standard deviation, and Root Mean Square (RMS).

Mean value of sensor readings is calculated using:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i \quad (1)$$

where μ represents the average sensor value, x_i represents the individual sensor sample, and N represents the total number of samples.

Standard deviation is calculated as:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (2)$$

where σ represents the variation in sensor readings.

Root Mean Square (RMS) value is calculated as:

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (3)$$

where RMS represents the effective magnitude of the sensor signal.

These extracted features improve the capability of the machine learning models to identify machine degradation and abnormal operating conditions.

3.5 Machine Learning Inference Layer

The processed feature vectors are passed to multiple machine learning models for fault classification, anomaly detection, and predictive analysis. The proposed framework utilizes a hybrid machine learning architecture consisting of Random Forest, XGBoost, Isolation Forest, and Long Short-Term Mem-

ory (LSTM) models(Taşcı et al., 2023),(Elkateb et al., 2024),(Rakholia et al., 2025).

Random Forest and XGBoost algorithms are used for supervised fault classification due to their high classification accuracy, robustness against noisy sensor data, and fast inference capability. These models classify machine conditions into normal and faulty operating states based on extracted sensor features.

Isolation Forest is implemented for unsupervised anomaly detection(Cummins et al., 2024). The model identifies abnormal operational patterns and detects outlier conditions that deviate from normal machine behavior. This enables early-stage anomaly detection even in cases where labeled fault data is limited.

The LSTM neural network model is used for time-series forecasting and Remaining Useful Life (RUL) estimation(Taşcı et al., 2023). Since industrial sensor data exhibits temporal dependencies, the LSTM model effectively captures sequential machine behavior and predicts future degradation trends.

The machine learning models were trained using historical sensor datasets collected under both normal and faulty machine conditions. The collected dataset included vibration, temperature, voltage, current, and RPM measurements generated during continuous machine operation. The dataset was divided into training and testing sets using an 80:20 split ratio.

Table 4 summarizes the machine learning model parameters used in the proposed system.

Table 4: Machine Learning Model Parameters

Parameter	Value
Fault Classification Models	Random Forest, XGBoost
Anomaly Detection Model	Isolation Forest
Time-Series Prediction Model	LSTM
Number of Features	18
Training Dataset Size	10000 Samples
Training Split	80%
Testing Split	20%
Validation Method	5-Fold Cross Validation

The machine learning inference module generates outputs including fault classification results, anomaly scores, predicted machine conditions, future degradation trends, and Remaining Useful Life estimations.

3.6 Cloud Database and Data Synchronization

Firebase Realtime Database is utilized as the cloud storage and synchronization platform within the proposed architecture. Firebase stores real-time sensor readings, processed datasets, machine learning out-

puts, anomaly scores, and predicted machine health information.

The edge computing node continuously uploads both live sensor data and machine learning prediction results to Firebase using REST API requests. The cloud database acts as a centralized storage platform enabling remote monitoring, synchronized data access, and future dataset retrieval for model retraining purposes.

The cloud storage structure is logically divided into the following categories:

- Real-Time Sensor Data Storage
- Processed Data Storage
- ML Prediction Storage
- Historical Fault Logs
- Remaining Useful Life Records

This layered storage structure improves data organization, scalability, and retrieval efficiency for industrial analytics applications.

3.7 Dashboard and Visualization Layer

The dashboard layer provides real-time visualization and remote monitoring of industrial machine conditions. The web dashboard application is developed locally using Python-based technologies and continuously retrieves live sensor readings and machine learning predictions from Firebase Realtime Database through REST API communication.

The dashboard supports dynamic real-time updates without requiring webpage refresh. Various visualization components including live charts, gauges, health indicators, status panels, and trend graphs are integrated to improve operator understanding of machine conditions.

The dashboard provides the following functionalities:

- Real-time sensor monitoring
- Historical trend visualization
- Fault prediction display
- Machine health estimation
- Remaining Useful Life visualization
- Alert and notification generation
- Anomaly status indication

Whenever abnormal machine conditions are detected, the system automatically generates alerts and notifications indicating the machine ID, fault severity, predicted failure type, and maintenance recommendation. This enables maintenance teams to take

corrective actions before catastrophic machine failure occurs.

The proposed architecture provides a scalable, low-cost, and real-time Industrial IoT predictive maintenance solution suitable for practical industrial deployment and smart manufacturing environments.

4 RESULT & DISCUSSION

4.1 Real-Time Sensor Monitoring Analysis

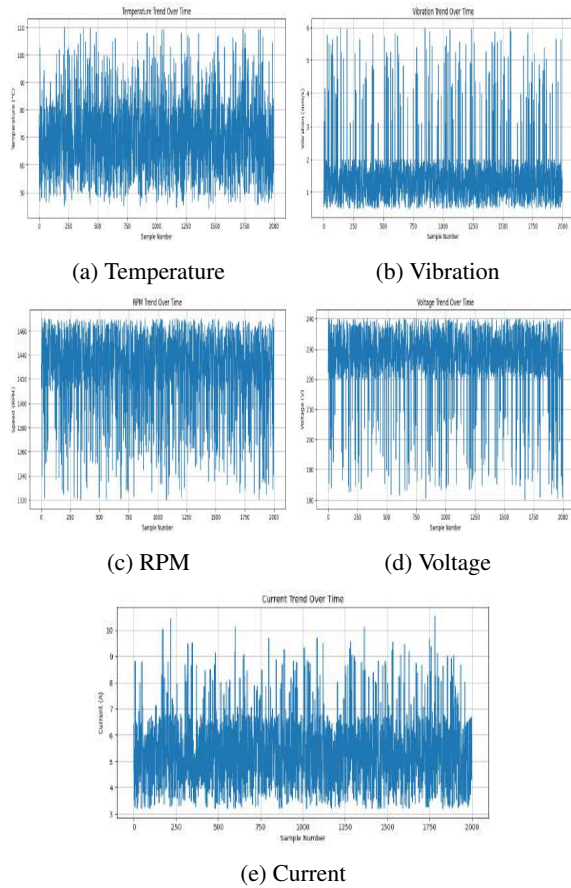


Figure 3: Real-time sensor monitoring results obtained from (a) Temperature, (b) Vibration, (c) RPM, (d) Voltage, and (e) Current sensors

The proposed predictive maintenance system was evaluated using real-time sensor data collected from temperature, vibration, RPM, voltage, and current sensors. Approximately 2000 samples were analyzed to monitor machine behavior and identify operational variations.

Temperature measurements varied between 45°C

and 110°C, where occasional spikes indicated possible heating effects caused by changing load conditions. Vibration values were generally maintained within 0.5 to 2 mm/s, while sudden peaks up to 6 mm/s suggested abnormal operating conditions such as bearing wear or shaft misalignment.

RPM readings ranged from 1320 to 1470 RPM, demonstrating stable machine operation with minor fluctuations during varying load conditions. Voltage levels were mostly maintained between 220 to 240 V, with small variations indicating minor power supply disturbances. Similarly, current consumption ranged from 3 to 10.5 A, where increased current values represented higher motor loading conditions.

Overall, the multi-sensor monitoring framework successfully captured real-time machine behavior and demonstrated effective continuous condition monitoring capability for predictive maintenance applications.

4.2 Machine Learning Model Performance Analysis

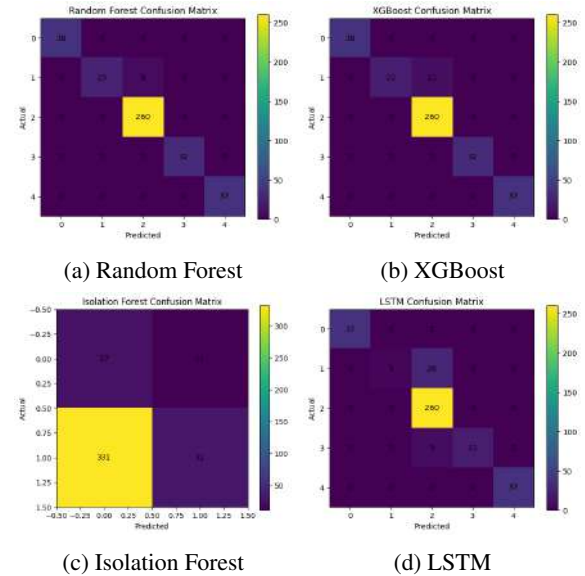


Figure 4: Confusion matrices of (a) Random Forest, (b) XGBoost, (c) Isolation Forest, and (d) LSTM models

The confusion matrices presented in Figure 4 demonstrate the classification performance of the implemented machine learning models for industrial fault detection. The Random Forest and XGBoost models achieved high classification accuracy with minimal misclassification errors, indicating effective fault prediction capability. The Isolation Forest model successfully identified abnormal operating conditions with fewer false detections, making it

suitable for anomaly detection tasks. Similarly, the LSTM model effectively captured temporal variations in machine behavior and provided reliable prediction performance.

The higher concentration of values along the diagonal regions of the confusion matrices indicates accurate classification results and validates the effectiveness of the proposed predictive maintenance framework for real-time industrial monitoring.

Table 5 presents the comparative performance analysis of the implemented machine learning models using accuracy, precision, recall, F1-score, and inference latency metrics.

Table 5: Performance Comparison of ML Models

Model	Accuracy	Precision	Recall	F1-Score	Latency
RF	94.2	93.8	94.5	94.1	320
XGB	93.4	92.9	93.7	93.3	280
IF	88.1	87.4	86.9	87.1	190
LSTM	93.1	92.5	92.8	92.6	1180

Overall, the Random Forest model achieved the best balance between classification accuracy and computational efficiency, while XGBoost provided reliable prediction performance with lower inference latency. Although the LSTM model demonstrated effective temporal learning capability, it required higher processing time due to sequential data analysis.

4.3 Real-Time Dashboard and System Visualization

The proposed Nirikshak framework includes a web-based real-time monitoring dashboard for continuous visualization of industrial machine parameters. Figures 5 and 6 present the developed interface integrated with machine learning-based fault prediction and cloud synchronization.

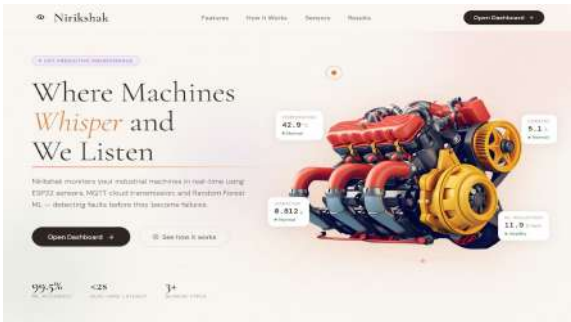


Figure 5: Web-based interface of the proposed Nirikshak framework

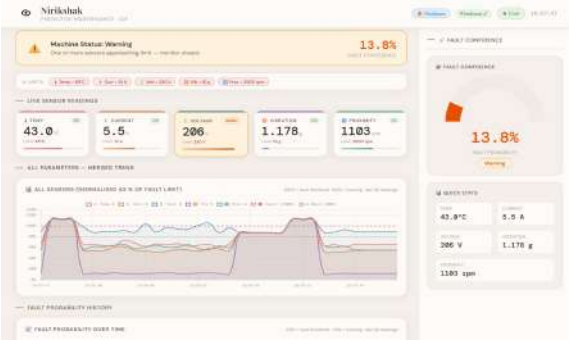


Figure 6: Real-time monitoring dashboard with fault analysis

The dashboard continuously displays temperature, vibration, voltage, current, and RPM sensor readings along with real-time fault probability estimation. In addition, merged trend visualization and warning indicators enable operators to identify abnormal machine behavior at an early stage. The integration of MQTT communication and Firebase cloud synchronization provides continuous remote monitoring capability with low-latency data updates.

The developed visualization framework improves operator awareness, enables proactive maintenance decision-making, and enhances the practical usability of the proposed predictive maintenance system in industrial environments.

4.4 Cost Analysis and Industrial Feasibility

Table 6: Estimated Hardware Cost Analysis

Component	Cost (INR)
STM32 Microcontroller	Rs.700
ESP8266 WiFi Module	Rs.400
Temperature Sensor	Rs.300
Vibration Sensor	Rs.1,000
RPM Sensor	Rs.500
Voltage and Current Sensors	Rs.900
PCB and Connectors	Rs.1,500
Power Supply	Rs.1,200
Miscellaneous Components	Rs.2,000
Total Hardware Cost	Rs.8,500

The proposed Nirikshak framework was developed using low-cost embedded hardware and open-source technologies. The total estimated implementation cost was approximately **Rs.8,500**, which is significantly lower than conventional predictive maintenance systems. The reduced implementation cost, low deployment complexity, and real-time monitoring ca-

pability make the proposed system economically feasible for small and medium-scale industries (SMEs). Furthermore, the modular architecture enables easy scalability for future industrial expansion and additional sensor integration.

5 CONCLUSIONS

The proposed *Nirikshak* framework presents an Industrial IoT-based predictive maintenance system designed for real-time machine monitoring and early fault detection. The framework integrates multi-sensor data acquisition, STM32-based edge processing, MQTT communication, machine learning inference, Firebase cloud synchronization, and real-time dashboard visualization into a unified architecture.

During the experiments conducted, the method proved highly efficient with a 94.2% rate of accurate fault classifications, with the latency for data transfer measured to be about 320 milliseconds. Moreover, the models were capable of detecting anomalies and deterioration in a timely manner before a catastrophic failure occurred. Furthermore, the system's construction cost only Rs. 8,500, which is a rather low price compared to that for other commercially available platforms.

As such, it can be argued that the presented system is a reliable proof-of-concept demonstrating how advanced predictive maintenance technology can be implemented using low-cost embedded hardware and open-source software. It is scalable, modular, and versatile enough to be implemented across many kinds of industries.

Future research directions include working on lightweight edge AI models, adaptive thresholding for better efficiency, online machine learning model retraining, improvements in cybersecurity measures, and implementation in large-scale industrial environments comprising multiple machines.

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- Recent and relevant references (preferably from the last 3–5 years) should be added to strengthen the literature review
- The revised paper need to submit after the registration process and instruction for the same will be shared on WhatsApp group of register author

Editorial Comments:-

We request you to complete the registration process by paying fees within 5 working days from the date of this email. Account details are mentioned in google form.

Registration and Payment Link:- <https://forms.gle/jfPDktV41BVZXabX6>

Fee details: <https://icaimest.in/Registration.php>

After registration further guidelines will be shared on WhatsApp group and link for the same is mentioned in last section of google form.

Once again, congratulations on your acceptance and thank you for your valuable contribution.

Regards,

Program Chair

ICAIMEST 2026

For Any Query Contact : 8446485823 / 9403670736

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